Collaborative Beam Hopping of Load Balancing and Interference Avoidance for Multi-GEO Satellite Systems Using QMIX

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***Abstract*—Beam hopping (BH) is a pivotal technology for the dynamic reconfiguration of time-space-frequency resources in multi-beam satellite (MBS) systems, significantly enhancing the on-orbit efficiency and flexibility of payloads. However, the increasing demand for satellite communications leads to high concurrency of multi-tasking and potential single-satellite overload under limited onboard payload resources. Moreover, co-frequency interference (CFI) resulting from overlapping beam illumination and bandwidth reuse adversely affects communication reliability and stability. To address these overlooked issues, this paper proposes a collaborative multi-satellite BH (CMSBH) method for joint load balancing and interference avoidance. Specifically, we formulate the problem as a multi-objective optimization of throughput, delay fairness, and load balancing. Additionally, a novel multi-level interference penalty mechanism is introduced to handle intra-satellite, inter-satellite, and neighboring cell CFI. We then employ a centralized training and decentralized execution (CTDE) framework using the QMIX model to learn a globally coordinated policy suitable for lightweight onboard deployment. The effectiveness of the proposed method is verified through simulation experiments.**

***Index Terms*—Beam hopping, multi-beam satellite, multi-satellite collaborative, interference avoidance, multi-agent deep reinforcement learning.**

I. Introduction

BEAM hopping (BH) is a critical technique for dynamically reconfiguring satellite resources and functionalities, enhancing the on-orbit efficiency and flexibility of the payload [1]. It controls the spatial orientation of satellite beams through the phased array antennas and employs the ideas of time division multiplexing (TDM) and frequency division multiplexing (FDM) to enables multiple access to temporal and spectral resources, thereby promoting efficient utilization of "time-space-frequency" resources [2], [3]. At present, DVB-RCS2/S2X standards [4] have proposed various protocols and specifications to support BH technology, indicating its significant potential in the context of space-air-ground integrated networks (SAGINs) [5]. Furthermore, related multi-beam satellite (MBS) systems equipped with BH technology are actually and widely used, especially the geostationary earth orbit (GEO) high-throughput satellite (HTS) system, such as Viasat [6], Inmarsat-5F [7], etc.

Against the above background, the motivations for this work are as follows: (1) **Multi-satellite collaborative and load-balanced BH:** for the continuously growing communication demands, under the condition of multi-task high-concurrency, the complexity of data collection and analysis by satellite payloads continues to rise, which can easily lead to single satellite overload. Taking advantage of clustering through multi-satellite collaboration to solve tasks beyond the capabilities of an individual satellite is a promising technology that provides more ubiquitous communication capabilities [8]. In addition, multi-satellite collaboration can also expand communication coverage and improve system resource utilization. However, the majority of current research efforts in BH still concentrate on the single-satellite system [1], [2], [9], [10], [11], [12], [13], [14], [15], [16]. (2) **Multi-satellite inter-beam interference avoidance BH**: moreover, the overlapping interference caused by beam radiation patterns and CFI caused by bandwidth reuse may lead to signal aliasing, increased bit error rate, and decreased communication efficiency, thereby affecting communications reliability and stability. However, most current works ignore this aspect [1], [2], [9], [10], [11], [13], [14], [15], [16]. According to work [17], when the spatial distance between beams exceeds a certain threshold, the interference effect can be negligible. Hence, precise spatial isolation between beams enabled by interference aware can promote the system to operate stably in hotspot areas. (3) **Multi-objective real-time and collaborative optimization**: finally, despite the existence of some MSBH methods [13], [18], [19], they generally decompose the problem into multiple sub-problems and optimize them independently. For instance, work [13] divides the problem into three sub-

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problems: cell planning, single satellite BH, and spatial isolation. This category of methods lacks real-time dynamic planning for the cell allocation during the MSBH process, and therefore cannot ensure real-time load balancing between satellites and real-time interference avoidance between beams. Therefore, multi-objective coordinated and real-time optimization are critical for providing effective MSBH decision-making.

Based on the above inspirations, we propose a collaborative MSBH (CMSBH) method with load balancing and interference avoidance. Specifically, the contributions are as follows,

1) A novel multi-satellite collaborative BH framework: To mitigate the single-satellite overload problem prevalent in existing systems, we propose a multi-satellite collaborative BH method that explicitly addresses inter-satellite load balancing. Among them, the optimization process is formulated to comprehensively consider the coordinated and real-time optimization of multiple objectives, including throughput, delay fairness, and load balancing.
2) A comprehensive multi-level interference avoidance model: to cope with the impact of overlapping interference caused by beam radiation directions and CFI caused by bandwidth reuse on communication stability and reliability, we design a novel interference penalty mechanism that systematically accounts intra-satellite beam interference, inter-satellite beam interference, and CFI from neighboring cells. This provides a more realistic and robust model for interference-aware resource allocation compared to prior works that often ignore or simplify interference constraints.
3) A centralized training and distributed execution (CTDE)-based MARL framework for cooperative optimization: to solve this complex multi-satellite, multi-objective optimization problem in real-time, we deploy a CTDE framework based on the QMIX model, which has ability to learn a globally coordinated policy while allowing for lightweight, decentralized execution on board. Extensive experimentation validated that the throughput of the most visible cells closely approximates its total traffic demand, with over 88.4% of cells achieving a traffic satisfaction rate exceeding 90%. In terms of load balancing, the load standard deviation (SD) of most cases are less than 12 Mbps. Furthermore, comparative experiments and ablation experiments have verified the effectiveness of the proposed method.

The subsequent sections are: Section II reviews the related works; Section III introduces the system modelling; Section IV presents the problem formulation; Section V describes the proposed multi-agent DRL model; In Section VI, extensive simulation experiments are conducted; Section VII summarizes this work.

II. Related Work

A. Traditional Heuristic-Based BH Methods

Early research on BH predominantly relied on heuristic algorithms. They use a series of empirical rules to quickly find approximate solutions to the problem. Typical strategies are based on instantaneous indicators to make decisions, such as traffic demand, quality of service (QoS), channel gain, etc. For instance, Zhang et al. [9] employed a genetic algorithm (GA) to optimize the beam hopping time plan (BHTP) considering rain attenuation. However, meta-heuristic approaches often suffer from high computational complexity and slow convergence, making them impractical for real-time applications, and this work lacks a multi-objective perspective that includes delay fairness or inter-satellite load balancing Chen et al. [1] proposed a BQP formulation for dynamic beam illumination, but solving the NP-hard BQP problem poses real-time challenges, and the model is inherently single-satellite centric, without mechanisms for inter-satellite collaboration. Han et al. [10] utilized a greedy algorithm to select cells with the maximum delay. This simple heuristic, while addressing delay fairness, may sacrifice overall system throughput due to its short-sighted nature. Similarly, Kibria et al. [14] proposed a joint BH and carrier aggregation (CA) scheme using mixed-integer nonlinear programming (MINLP). Although it improves user fairness, solving such a large-scale MINLP problem is computationally intensive and thus unsuitable for dynamic environments.

Overall, traditional heuristic-based methods are often too computationally demanding for real-time deployment, tend to find local optima in high-dimensional search spaces, and most critically, are limited to single-satellite scenarios, thus ignoring the vital aspects of inter-satellite collaboration and load balancing [20].

B. RL-Enabled BH Methods

In contrast, AI-based optimization algorithms, particularly deep reinforcement learning (DRL), are more promising for solving complex, sequential decision-making problems like BH [13], [21], [22]. For example, Hu et al. [11] modeled the entire satellite as a single agent using parallel deep Q-networks (DQN) to optimize a weighted sum of throughput, delay, and fairness objectives.. However, this single-agent formulation faces the 'curse of dimensionality,' as the action space grows combinatorially with the number of beams, making it intractable. Kim et al. [15] proposed a two-step DQN-based approach that first allocates time slots and then optimizes power. This sequential optimization, however, can lead to globally suboptimal performance as the two steps are not jointly optimized. Ran et al. [16] employed a parametric DQN (P-DQN) to handle the hybrid action space, but the architecture remains a centralized single-agent system with scalability concerns for a large number of beams. To overcome this, multi-agent DRL (MADRL) has emerged as a promising paradigm. Lin et al. [12] introduced a cooperative MADRL framework where each beam is modeled as an agent. This effectively decomposes the action space, but its coordination is confined to the intra-satellite level, lacking mechanisms for inter-satellite load balancing or interference management. Furthermore, Zhao et al. [2] proposed a delay-sensitive BH method based on fully decentralized soft actor-critic (SAC) by clustering beams. Among them, each cluster is managed

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TABLE I
COMPARISON OF RELATED WORKS

Work	Method Basis	Key Focus	Limitations in Multi-Goal Coordination & Real-Time Performance
[9]	GA	Optimizes BHTP and spectrum allocation to maximize capacity.	High computational complexity and slow convergence of GA are unsuitable for real-time decision-making. The framework is primarily single-objective and lacks explicit inter-satellite load balancing and delay fairness coordination.
[1]	BQP	Models beam illumination and selective precoding to minimize an interference penalty.	NP-hard BQP poses real-time challenges for large-scale systems. The model is single-satellite focused, lacking explicit mechanisms for inter-satellite load balancing and collaboration.
[10]	Greedy algorithm	Prioritizes cells with the highest delay to improve delay fairness.	The myopic (short-sighted) nature of the greedy strategy can sacrifice global throughput. It lacks a mechanism to coordinate between delay fairness, capacity, and inter-satellite load balancing.
[14]	MILP	Jointly optimizes BH and CA to improve user fairness.	Computationally expensive MINLP/MILP formulation is not viable for real-time operation. The approach is limited to a single satellite, neglecting inter-satellite load balancing and interference management.
[11]	DQN	Models a single satellite as an agent to optimize throughput and delay fairness.	Suffers from the curse of dimensionality as the action space grows combinatorially. The single-agent model is incapable of handling inter-satellite coordination, such as load balancing and interference management.
[15]	DQN+ Iterative Opt.	Decouples BH into time slot allocation and power allocation.	Sequential optimization is suboptimal compared to joint allocation. The framework remains single-satellite-centric and does not address multi-satellite collaborative load balancing or interference.
[16]	P-DQN	Solves the hybrid discrete-continuous action space of joint BH and power allocation.	The single-agent architecture has scalability concerns and is not designed for multi-satellite systems. It inherently lacks mechanisms for inter-satellite load balancing and collaboration.
[12]	MADDQN	Assigns one agent per beam to decompose the action space.	Focuses on intra-satellite coordination, neglecting inter-satellite interference and load balancing, which are critical in multi-satellite systems.
[2]	Decentralized MASAC	Decomposes BH into independent intra-cluster problems, each solved by a local SAC agent.	The lack of explicit coordination mechanisms risks globally suboptimal policies, as agents act greedily on local information. This is ineffective for system-wide objectives like inter-satellite load balancing that require global state awareness.
CMSBH	Cooperative MARL (QMIX)	Proposes a collaborative multi-satellite BH framework for joint load balancing and interference avoidance.	Employs a CTDE paradigm to explicitly learn collaborative policies for inter-satellite load balancing, moving beyond single-satellite-centric models; Simultaneously optimizes throughput, delay fairness, and load balancing in an end-to-end manner, avoiding suboptimal sequential decision-making; Incorporates a multi-level penalty mechanism for intra-satellite, inter-satellite, and co-frequency interference.

by a SAC agent responsible for beam allocation and resource allocation within the cluster. While this offers excellent scalability, it lacks an explicit coordination mechanism between agents, risking convergence to suboptimal states, especially in multi-satellite systems where global coordination is critical for load balancing.

Collectively, a common thread across most existing works, from traditional optimization [1], [9], [10], [14] to advanced DRL frameworks [2], [11], [12], [15], [16], is their inherent single-satellite-centric design. They consequently lack the mechanisms for explicit inter-satellite coordination, rendering them unable to address system-level challenges like dynamic load balancing and interference avoidance in a multi-satellite constellation. Furthermore, even in multi-satellite BH (MSBH) methods like [13], the problem is typically decomposed into sequential subproblems. This staged optimization sacrifices real-time, multi-objective coordination, as decisions made in one stage can lead to globally suboptimal outcomes in subsequent stages. Crucially, a unified framework for simultaneous, real-time, joint optimization of throughput, delay fairness, load balancing, and multi-level interference remains an open challenge. To fill these gaps, this paper proposes a CMSBH method leveraging a CTDE paradigm with QMIX to learn a globally coherent policy for multi-satellite collaboration, multi-objective real-time optimization, and systematic interference management. The detailed comparison is summarized in Table I.

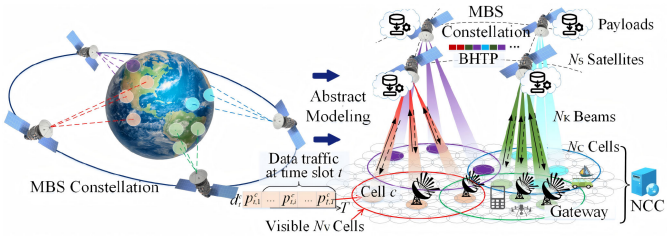


Fig. 1. The MBS constellation communication system architecture.

III. SYSTEM MODELING

A. The MBS Constellation Communication System

As illustrated in Fig. 1, the MBS constellation communication system comprises both the space and ground segments. Specifically, N^{s} satellites positioned at an orbital height H collaboratively serve N^{c} cells, with each satellite equipped with N^{k} beams covering N^{v} visible cells ($N^{\mathsf{v}} < N^{\mathsf{c}}$). Therefore, the sets of satellites, beams, cells, and visible cells are respectively represented as $\mathbb{S} = \{s \mid 1, 2, \cdots, N^{\mathsf{s}}\}$, $\mathbb{K} = \{k \mid 1, 2, \cdots, N^{\mathsf{k}}\}$, $\mathbb{C} = \{c \mid 1, 2, \cdots, N^{\mathsf{c}}\}$, and $\mathbb{V} = \{v \mid 1, 2, \cdots, N^{\mathsf{v}}\}$. Furthermore, due to the coverage overlap of certain cells by multiple satellites, $\sum_{s=1}^{N^{\mathsf{s}}} N_s^{\mathsf{v}} \leq N^{\mathsf{c}}$ and $\cup_{s=1}^{N^{\mathsf{s}}} \mathbb{V}_s = \mathbb{C}$, which enables these cells to establish communication with multiple satellites. It is important to note that this is a significant characteristic of the multi-satellite

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BH. Moreover, for simplicity, we set equal N^{k} and N^{v} for each satellite, *i.e.*, $N_1^{\mathsf{k}} = \cdots = N_s^{\mathsf{k}} = \cdots = N_{N^{\mathsf{s}}}^{\mathsf{k}}$ and $N_1^{\mathsf{v}} = \cdots = N_s^{\mathsf{v}} = \cdots = N_{N^{\mathsf{s}}}^{\mathsf{v}}$. In practice, through this research, it can be easily extended to more dimensions and possible values of N^{k} and N^{v}.

At time slot t, the data traffic of all cells is denoted as $\boldsymbol{D}_t = \{\boldsymbol{d}_{c,t} \mid c \in \mathbb{C}\}$, where $\boldsymbol{d}_{c,t}$ denotes the data traffic of cell c at time slot t. Let $p_{c,t,i}$ denote the data packets that have been waiting for i time slots in $\boldsymbol{d}_{c,t}$. Thus, $\boldsymbol{d}_{c,t} = \{p_{c,t,1}, \cdots, p_{c,t,i}, \cdots, p_{c,t,T}\}$, where T represents the packet's lifetime and indicates packets that waiting for more than T time slots will be discarded. Similarly, the data traffic $\boldsymbol{D}_{s,t}$ of visible cells of the satellite s is expressed as $\boldsymbol{D}_{s,t} = \{\boldsymbol{d}_{s,v,t} \mid v \in \mathbb{V}_s\}$, where $\boldsymbol{d}_{s,v,t} = \{p_{s,v,t,1}, \cdots, p_{s,v,t,i}, \cdots, p_{s,v,t,T}\}$.

It should be noted that the latest arriving data traffic $q_{s,v,t}$ of cell v covered by satellite s at time slot t is generated according to Poisson distribution, as

$$P(q_{s,v,t} = n) = \frac{(\lambda_{s,v,t})^n \times e^{-\lambda_{s,v,t}}}{n!}, \tag{1}$$

where $\lambda_{s,v,t}$ represents the traffic arrival rate of cell v covered by satellite s at time slot t. Therefore, the latest arriving date traffic of all cells at time slot t can be expressed as $\boldsymbol{q}_t = [q_{1,t}, q_{2,t}, \cdots, q_{c,t}, \cdots, q_{N^{\mathsf{c}},t}]^{\mathrm{T}}$.

B. Beam Hopping

Focusing on the downlink, this research explores the BH strategy for the MBS constellation-terrestrial integrated communication system, whose abstract modeling is illustrated in Fig. 1. Based on the phased-array antennas and TDM technologies, it can be found that the BH problem for the MBS constellation can be defined as the task of inferring the beam's illumination direction and occupied spectrum resources for each satellite during every time slot based on the ground traffic demands, which is called beam allocation and spectrum allocation, as shown in Eq. (2a) and Eq. (3a), respectively. Among them, the spectrum resources of satellites directly impact the transmission capabilities of beams [23]. Constrained volume, weight, and energy consumption of on-board equipment, the rational, efficient, and flexible allocation of spectrum resources become crucial for improving the operational efficiency of the MBS constellation.

1) **Beam allocation**: At time slot t, which N^{k} cells are illuminated within the visible coverage of each satellite, as

$$\begin{aligned}\boldsymbol{\Xi}_t &= \left[\boldsymbol{\xi}_{1,t}, \boldsymbol{\xi}_{2,t}, \cdots, \boldsymbol{\xi}_{s,t}, \cdots, \boldsymbol{\xi}_{N^{\mathsf{s}},t}\right]^{\mathrm{T}} \\ &= \begin{bmatrix} \xi_{1,1,t} & \xi_{1,2,t} & \cdots & \xi_{1,k,t} & \cdots & \xi_{1,N_1^{\mathsf{k}},t} \\ \xi_{2,1,t} & \xi_{2,2,t} & \cdots & \xi_{2,k,t} & \cdots & \xi_{2,N_2^{\mathsf{k}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \xi_{s,1,t} & \xi_{s,2,t} & \cdots & \xi_{s,k,t} & \cdots & \xi_{s,N_s^{\mathsf{k}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \xi_{N^{\mathsf{s}},1,t} & \xi_{N^{\mathsf{s}},2,t} & \cdots & \xi_{N^{\mathsf{s}},k,t} & \cdots & \xi_{N^{\mathsf{s}},N_{N^{\mathsf{s}}}^{\mathsf{k}},t} \end{bmatrix}, \end{aligned} \tag{2a}$$

$$\text{s.t. } \boldsymbol{\xi}_{s,k,t} = [\xi_{s,k,1,t}, \cdots, \xi_{s,k,v,t}, \cdots, \xi_{s,k,N_s^{\mathsf{v}},t}], \quad \forall s \in \mathbb{S}, \forall k \in \mathbb{K}_s, \tag{2b}$$

$$\xi_{s,k,v,t} = \{0,1\}, \forall s \in \mathbb{S}, \forall k \in \mathbb{K}_s, \forall v \in \mathbb{V}_s, \tag{2c}$$

$$\sum_{k=1}^{N^{\mathsf{k}}} \sum_{v=1}^{N_s^{\mathsf{v}}} \xi_{s,k,v,t} = N^{\mathsf{k}}, \ \forall s \in \mathbb{S}, \forall k \in \mathbb{K}_s, \tag{2d}$$

$$\sum_{s=1}^{N^{\mathsf{s}}} \sum_{k=1}^{N^{\mathsf{k}}} \xi_{s,k,c,t} = 1, \ \forall c \in \mathbb{C}, \tag{2e}$$

$$\xi_{s,k,c',t} = 0, \ \forall c' \notin \mathbb{V}_s, \tag{2f}$$

where $\boldsymbol{\xi}_{s,t}$ represents the beam allocation pattern of satellite s; Constraint (2b) implies the illumination direction for beam k of satellite s at time slot t; Constraint (2c) implies that ξ is a binary variable, where $\xi_{s,k,v,t} = 1$ implies the cell v within the coverage of satellite s is illuminated by beam k, otherwise, $\xi_{s,k,v,t} = 0$ implies the cell v is not illuminated by satellite s; Constraint (2d) implies that each satellite activates N^{k} beams and serves N^{k} distinct cells at the same time; Constraint (2e) implies that each cell c at each time slot t is only permitted to be served by one beam, *i.e.*, the correspondence between the served cell and the serving beam is $1:1$; Constraint (2f) implies that only cells within visible coverage of the satellite can be illuminated.

2) **Spectrum allocation**: Based on the idea of FDM, we assume that the total bandwidth resources of each satellite are divided into multiple sub-frequency bands to improve spectrum utilization. Different beams can select the optimal bandwidth scheme required from a variety of different reuse schemes to flexibly allocate spectrum resources. In addition, we also require that each beam is only permitted to occupy the continuous frequency band to ensure that the transmitted signal has the carrier with a larger bandwidth, thereby achieving higher communication quality.

For each satellite s, the total bandwidth is B_s^{tot} and the number of sub-frequency bands is N_s^{sub}, then the allocated bandwidth for each sub-frequency band is $B_s^{\mathsf{sub}} = B_s^{\mathsf{tot}}/N_s^{\mathsf{sub}}$. Therefore, it can be deduced that there are $N_s^{\mathsf{reu}} = N_s^{\mathsf{sub}} \times (N_s^{\mathsf{sub}} + 1)/2$ reuse schemes in total. A sample of the bandwidth reuse schemes in five sub-frequency bands is illustrated in Fig. 2. Therefore, at time slot t, the spectrum allocation for each beam is expressed as

$$\begin{aligned}\boldsymbol{\Phi}_t &= \left[\boldsymbol{\phi}_{1,t}, \boldsymbol{\phi}_{2,t}, \cdots, \boldsymbol{\phi}_{s,t}, \cdots, \boldsymbol{\phi}_{N^{\mathsf{s}},t}\right]^{\mathrm{T}} \\ &= \begin{bmatrix} \phi_{1,1,t} & \phi_{1,2,t} & \cdots & \phi_{1,k,t} & \cdots & \phi_{1,N_1^{\mathsf{k}},t} \\ \phi_{2,1,t} & \phi_{2,2,t} & \cdots & \phi_{2,k,t} & \cdots & \phi_{2,N_2^{\mathsf{k}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \phi_{s,1,t} & \phi_{s,2,t} & \cdots & \phi_{s,k,t} & \cdots & \phi_{s,N_s^{\mathsf{k}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \phi_{N^{\mathsf{s}},1,t} & \phi_{N^{\mathsf{s}},2,t} & \cdots & \phi_{N^{\mathsf{s}},k,t} & \cdots & \phi_{N^{\mathsf{s}},N_{N^{\mathsf{s}}}^{\mathsf{k}},t} \end{bmatrix}, \end{aligned} \tag{3a}$$

$$\text{s.t. } \phi_{s,k,t} = \{1, 2, \cdots, N_s^{\mathsf{reu}}\}, \forall s \in \mathbb{S}, \ \forall k \in \mathbb{K}_s, \tag{3b}$$

$$|\phi_{s,k,t}| \times B_s^{\mathsf{sub}} \leq B_s^{\mathsf{tot}}, \forall s \in \mathbb{S}, \ \forall k \in \mathbb{K}_s, \tag{3c}$$

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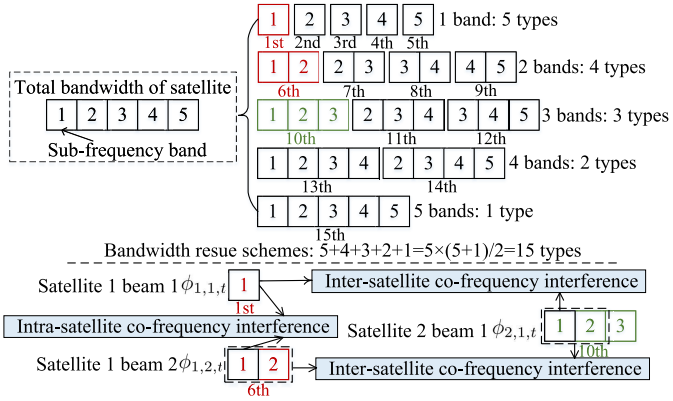


Fig. 2. A sample of the bandwidth reuse schemes in five sub-frequency bands.

where $\boldsymbol{\phi}_{s,t}$ represents the spectrum allocation pattern of satellite s; $\phi_{s,k,t}$ represents which bandwidth reuse scheme is chosen for beam k of satellite s; Constraint (3b) implies that the optional range of bandwidth reuse schemes for beams; Constraint (3c) implies the bandwidth occupied by each beam does not exceed the total bandwidth, where $|\phi_{s,k,t}|$ represents the number of sub-frequency bands occupied by beam k of satellite s.

Furthermore, it should be noted that different bandwidth reuse schemes chosen by beams from the same/different satellites may involve the same sub-frequency bands, inevitably leading to intra-satellite/inter-satellite beam CFI. For instance, in Fig. 2, beam 1 and beam 2 of satellite 1 choose the 1st ($\phi_{1,1,t} = 1$) and 6th ($\phi_{1,2,t} = 6$) bandwidth reuse schemes respectively, while beam 1 of satellite 2 chooses the 10th ($\phi_{2,1,t} = 10$) bandwidth reuse scheme. Consequently, there will be intra-satellite CFI between $\phi_{1,1,t}$ and $\phi_{1,2,t}$, and inter-satellite CFI between $\phi_{1,1,t}$ and $\phi_{2,1,t}$ as well as $\phi_{1,2,t}$ and $\phi_{2,1,t}$. CFI among beams will directly impact the Signal-to-Interference-plus-Noise Ratio (SINR) and channel capacity of the wireless channel.

C. Channel Modeling

Based on the Shannon formula, the channel capacity $C_{s,k,v,t}$ for the visual cell v illuminated by beam k of satellite s at time slot t is calculated as

$$C_{s,k,v,t} = \xi_{s,k,v,t} \times B_{s,k,t} \times \log\left(1 + \Gamma_{s,k,v,t}\right), \tag{4}$$

where $B_{s,k,t}$ represents the bandwidth occupied by beam k of satellite s, and its calculation is as

$$B_{s,k,t} = |\phi_{s,k,t}| \times B_s^{\mathsf{sub}}. \tag{5}$$

Moreover, $\Gamma_{s,k,v,t}$ represents the SINR for the visual cell v illuminated by beam k of satellite s at time slot t, as

$$\Gamma_{s,k,v,t} = \frac{P^{\mathsf{s}}_{s,k,v,t}}{P^{\mathsf{i}}_{s,k,v,t} + P^{\mathsf{n}}_{s,k,v,t}}, \tag{6}$$

where $P^{\mathsf{s}}_{s,k,v,t}$, $P^{\mathsf{i}}_{s,k,v,t}$, and $P^{\mathsf{n}}_{s,k,v,t}$ represent the signal power, interference power, and noise power, respectively. Their calculation methods are as

$$P^{\mathsf{s}}_{s,k,v,t} = \omega_{s,k} \times h_{s,k,v,t},$$

$$\text{s.t.} \quad \omega_{s,k} \le P^{\mathsf{max}}, \forall s \in \mathbb{S}, \ \forall k \in \mathbb{K}_s, \tag{7a}$$

$$\sum_{k=1}^{N_s^{\mathsf{k}}} \omega_{s,k} \le P^{\mathsf{tot}}, \forall s \in \mathbb{S}, \tag{7b}$$

where $\omega_{s,k}$ represents the transmit power for beam k of satellite s; Constraint (7a) implies that for each satellite s, the transmit power for each beam k does not exceed the maximum power limit P^{max}; Constraint (7b) implies that for each satellite s, the sum of the transmit power of all beams does not exceed the total power P^{tot} of the satellite; $h_{s,k,v,t}$ represents the channel coefficient for the visual cell v illuminated by beam k of satellite s at time slot t, and its calculation is as

$$h_{s,k,v,t} = G^{\mathsf{tran}}_{s,k,v} \times P^{\mathsf{L}} \times G^{\mathsf{rec}}_{s,k,v}, \tag{8}$$

where $G^{\mathsf{tran}}_{s,k,v}$ represents the transmit antenna gain for the beam k of satellite s; $G^{\mathsf{rec}}_{s,k,v}$ represents the receive antenna gain for the visual cell v illuminated by beam k of satellite s. Since the values of $G^{\mathsf{tran}}_{s,k,v}$ and $G^{\mathsf{rec}}_{s,k,v}$ are related to the specific simulation scenario, their calculation methods will be explained in detail in Section VI; P^{L} represents the free space signal propagation path loss, and its calculation is as

$$P^{\mathsf{L}} = 20 \times \log_{10}(H) + 20 \times \log_{10}(f^{\mathsf{ka}}) + 32.44, \tag{9}$$

where f^{ka} represents the downlink frequency, and in this study, the Ka-band is utilized, which will be explained in Section VI.

$$P^{\mathsf{i}}_{s,k,v,t} = P^{\mathsf{i,inter}}_{s,k,v,t} + P^{\mathsf{i,intra}}_{s,k,v,t}, \tag{10}$$

where $P^{\mathsf{i,inter}}_{s,k,v,t}$ represents the inter-satellite CFI power; $P^{\mathsf{i,intra}}_{s,k,v,t}$ represents the intra-satellite CFI power, as

$$P^{\mathsf{i,inter}}_{s,k,v,t} = \sum_{s'=1}^{N^{\mathsf{s}}} \sum_{k'=1}^{N_{s'}^{\mathsf{k}}} \sum_{v'=1}^{N_{s'}^{\mathsf{v}}} \left(\xi_{s',k',v',t} \cdot \omega_{s',k'} \cdot h_{s',k',v',t} \cdot \sigma_{k',k,t}\right), \quad \forall s' \ne s, \forall k' \in \mathbb{K}_{s'}, \forall v' \in \mathbb{V}_{s'}, v' \ne v, \tag{11}$$

$$P^{\mathsf{i,intra}}_{s,k,v,t} = \sum_{k'=1}^{N_s^{\mathsf{k}}} \sum_{v'=1}^{N_s^{\mathsf{v}}} \left(\xi_{s,k',v',t} \cdot \omega_{s,k'} \cdot h_{s,k',v',t} \cdot \sigma_{k',k,t}\right), \quad \forall k' \in \mathbb{K}_s, \forall v' \in \mathbb{V}_s, v' \ne v, \tag{12}$$

where $v' \ne v$ in Eq. (11) implies a one-to-one relationship between the cell in the overlapping region of satellite coverage and the beam; $\sigma_{k',k,t}$ represents the frequency overlap factor between beam k' and beam k, denoting the CFI from beam k' to beam k resulting from the bandwidth reuse schemes selected by different beams, and its calculation is as

$$\sigma_{k',k,t} = \frac{|\phi_{s',k',t} \cap \phi_{s,k,t}|}{|\phi_{s',k',t}|}, \tag{13}$$

where $|\phi_{s',k',t} \cap \phi_{s,k,t}|$ represents the number of overlapping sub-frequency bands occupied by beam k' and beam k.

$$P^{\mathsf{n}}_{s,k,v,t} = N_0 \times B_{s,k,t}, \tag{14}$$

where N_0 represents the Noise power spectral density.

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IV. Problem Formulation

The BH problem can be modeled as: under the limited resource constraints of the MBS constellation, the data transmission capacity (Throughput) is optimized as much as possible to meet the ground traffic demand, while ensuring delay fairness (Delay Offset) for ground users and load balancing (Load Offset) for each satellite.

A. Optimize Transmission Capacity by Maximizing Throughput

The throughput for the visual cell v illuminated by beam k of satellite s at time slot t is calculated as

$$\mathcal{T}_{s,v,k,t} = \min\left\{C_{s,k,v,t}, \sum_{i=1}^{T} p_{s,k,v,t,i}\right\}. \tag{15}$$

The overall throughput for the constellation is calculated as

$$\mathcal{T}_t^{\mathsf{all}} = \sum_{s=1}^{N^{\mathsf{s}}} \sum_{k=1}^{N_s^{\mathsf{k}}} \sum_{v=1}^{N_s^{\mathsf{v}}} \left(\xi_{s,k,v,t} \times \mathcal{T}_{s,v,k,t}\right). \tag{16}$$

Furthermore, it is essential to focus on the long-term throughput, which is calculated as

$$\mathcal{T}^{\mathsf{lt}} = \sum_{t=1}^{+\infty} \mathcal{T}_t^{\mathsf{all}}. \tag{17}$$

Therefore, this optimization objective is modelled as

$$\begin{aligned} \text{opt. } \mathcal{F}_t^1 : \ & \max_{\boldsymbol{\Xi}_t, \boldsymbol{\Phi}_t} \mathcal{T}^{\mathsf{lt}}, \\ & \text{s.t. (2b)-(2f), (3b), (3c), (7a), (7b), (19a),} \\ & \quad \text{(24a), (24b),} \end{aligned} \tag{18}$$

where $\max_{\boldsymbol{\Xi}_t, \boldsymbol{\Phi}_t}$ indicates that beam and spectrum resources allocation variables are optimized; (19a), (24a), and (24b) are the delay and load balancing related constraints that should be met respectively, which will be introduced below.

B. Optimize Delay Fairness by Minimizing Delay Offset

Not only that, the BH decision-making process also needs to consider the fairness of delay differences between different users, defined as the delay offset. The objective is to ensure that no cell, regardless of its traffic characteristics, suffers from excessively long waiting times or potential starvation, thereby providing a baseline level of service fairness at the network resource allocation layer. We acknowledge that a more fine-grained service-oriented QoS fairness metric could incorporate the diverse service priorities of different traffic types (e.g., distinguishing between real-time and non-real-time services). Such a service-aware fairness model represents a valuable and more complex research direction. However, the scope of this work is to establish a robust and fundamental mechanism for equitable resource distribution across all competing cells. The proposed queuing delay fairness metric serves as a critical first-order approximation for overall user experience and is a foundational step upon which more sophisticated, service-differentiated scheduling policies could be built in future work.

At time slot t, the delay offset for the visible cells of each satellite is expressed as

$$\begin{aligned} \boldsymbol{\Delta}_t &= [\boldsymbol{\delta}_{1,t}, \boldsymbol{\delta}_{2,t}, \cdots, \boldsymbol{\delta}_{s,t}, \cdots, \boldsymbol{\delta}_{N^{\mathsf{s}},t}]^{\mathrm{T}} \\ &= \begin{bmatrix} \delta_{1,1,t} & \delta_{1,2,t} & \cdots & \delta_{1,v,t} & \cdots & \delta_{1,N_1^{\mathsf{v}},t} \\ \delta_{2,1,t} & \delta_{2,2,t} & \cdots & \delta_{2,v,t} & \cdots & \delta_{2,N_2^{\mathsf{v}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \delta_{s,1,t} & \delta_{s,2,t} & \cdots & \delta_{s,v,t} & \cdots & \delta_{s,N_s^{\mathsf{v}},t} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \delta_{N^{\mathsf{s}},1,t} & \delta_{N^{\mathsf{s}},2,t} & \cdots & \delta_{N^{\mathsf{s}},v,t} & \cdots & \delta_{N^{\mathsf{s}},N_{N^{\mathsf{s}}}^{\mathsf{v}},t} \end{bmatrix}, \\ \text{s.t. } \delta_{s,v,t} &= \max_{v' \in \mathbb{V}^s} \Delta_{s,v',t} - \min_{v'' \in \mathbb{V}^s} \Delta_{s,v'',t}, \\ & \quad \forall s \in \mathbb{S}, \forall v, v', v'' \in \mathbb{V}_s, \end{aligned} \tag{19a}$$

where $\boldsymbol{\delta}_{s,t}$ represents the delay allocation pattern of satellite s; $\delta_{s,v,t}$ represents the delay offset for the visible cell v of satellite s; Constraint (19a) implies that the delay offset is equal to the difference between the maximum and the minimum average queuing delay of data packets within the visual cells of its associated satellite. Here, $\Delta_{s,v',t}$ represents the average queuing delay of data packets for the visual cell v' of satellite s

$$\Delta_{s,v',t} = \frac{\sum_{i=1}^{T} (i \times p_{s,v',t,i})}{\sum_{i=1}^{T} p_{s,v',t,i}}. \tag{20}$$

The overall delay offset for the constellation is calculated as

$$\mathcal{D}_t^{\mathsf{all}} = \sum_{s=1}^{N^{\mathsf{s}}} \sum_{k=1}^{N_s^{\mathsf{k}}} \sum_{v=1}^{N_s^{\mathsf{v}}} \left(\xi_{s,k,v,t} \times \delta_{s,v,k,t}\right). \tag{21}$$

Likewise, it is essential to focus on the long-term delay offset, which is calculated as

$$\mathcal{D}^{\mathsf{lt}} = \sum_{t=1}^{+\infty} \mathcal{D}_t^{\mathsf{all}}. \tag{22}$$

Therefore, this optimization objective is modelled as

$$\begin{aligned} &\text{opt. } \mathcal{F}_t^2 : \ \min_{\boldsymbol{\Xi}_t, \boldsymbol{\Phi}_t} \mathcal{D}^{\mathsf{lt}}, \\ &\text{s.t. (2b)-(2f), (3b), (3c), (7a), (7b), (19a), (24a), (24b).} \end{aligned} \tag{23}$$

C. Optimize Load Balancing by Minimizing Load Offset

In addition, inter-satellite load offset needs to be considered, which is critical for load balancing on each satellite. The core principle is not to arbitrarily shift load between satellites, but rather to dynamically assign cells within these overlapping areas to the most suitable satellite at any given time. At time slot t, the load for each satellite is expressed as

$$\begin{aligned} \boldsymbol{\Lambda}_t &= [\lambda_{1,t}, \lambda_{2,t}, \cdots, \lambda_{s,t}, \cdots, \lambda_{N^{\mathsf{s}},t}], \\ \text{s.t. } \lambda_{s,0} &= 0, \forall s \in \mathbb{S}, \end{aligned} \tag{24a}$$

$$\lambda_{s,t} = \sum_{k=1}^{N_s^{\mathsf{k}}} \sum_{v=1}^{N_s^{\mathsf{v}}} \left(\xi_{s,k,v,t} \times \mathcal{T}_{s,v,k,t}\right), \forall s \in \mathbb{S}, \tag{24b}$$

where $\lambda_{s,t}$ represents the load of satellite s; Constraint (24a) represents the initial value of $\lambda_{s,t}$; Constraint (24b) implies

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that the load of the satellite is the total traffic of cells it has served.

The overall load offset for the constellation is calculated as

$$\mathcal{L}_t^{\mathsf{all}} = \lambda_{\arg\max_s \lambda_{s,t},t} - \lambda_{\arg\min_{s'} \lambda_{s',t},t}. \tag{25}$$

It should be noted that the goal of minimizing the load offset thus incentivizes the algorithm to make intelligent dynamic cell-to-satellite assignment decisions in these shared regions. For example, if a hotspot cell is visible to both a heavily loaded satellite and a lightly loaded one, the algorithm will learn to assign it to the latter to balance the load, thereby improving the overall joint capacity and resource utilization of the constellation. It encourages the system to operate as a coordinated whole, rather than a collection of independent entities.

Likewise, it is essential to focus on the long-term load offset, which is calculated as

$$\mathcal{L}^{\mathsf{lt}} = \sum_{t=1}^{+\infty} \mathcal{L}_t^{\mathsf{all}}. \tag{26}$$

Therefore, this optimization objective is modelled as

$$\begin{aligned} \text{opt. } \mathcal{F}_t^3 =: & \min_{\mathbf{\Xi}_t, \mathbf{\Phi}_t} \mathcal{L}^{\mathsf{lt}}, \\ & \text{s.t. (2b)-(2f), (3b),} \\ & \text{(3c), (7a), (7b), (19a), (24a), (24b).} \end{aligned} \tag{27}$$

In summary, combining Eq. (18), Eq. (23), and Eq. (27), the overall optimization goal is defined as

$$\begin{aligned} \text{opt. } & \max_{\mathbf{\Xi}_t, \mathbf{\Phi}_t} \mathcal{F}_t = \alpha^{\mathcal{T}} \times \mathcal{T}^{\mathsf{lt}} - \beta^{\mathcal{D}} \times \mathcal{D}^{\mathsf{lt}} - \iota^{\mathcal{L}} \times \mathcal{L}^{\mathsf{lt}}, \\ & \text{s.t. (2b)-(2f), (3b), (3c), (7a), (7b), (19a), (24a), (24b).} \end{aligned} \tag{28}$$

where $\alpha^{\mathcal{T}}$, $\beta^{\mathcal{D}}$, and $\iota^{\mathcal{L}}$ represent the rescaling factors for ensuring that different components are on the same scale, respectively.

D. Interference Avoidance by Setting Penalty Mechanism

To further clarify the operational motivation of the considered scenario, it should be noted that this work does not aim to represent a widely separated multi-GEO deployment. Instead, it focuses on a closely spaced adjacent-GEO operating pattern, where neighboring GEO satellites perform spectrum reuse over overlapping coverage areas. Regulatory documents of the International Telecommunication Union (ITU) explicitly state that orbit/spectrum resources, especially GEO resources, are limited natural resources and should be used in a rational, efficient, and economical manner [24]. Recent official ITU materials have also pointed out that GEO positions are becoming increasingly crowded, and the risk of inter-system interference continues to grow [25].

In addition, the compatibility and coordination of adjacent GEO systems with small angular separation has already been formally studied by the ITU. ITU-R S.2409 [26] discusses interference issues for closely separated GEO FSS/VSAT networks in the Ka band. ITU-R S.2464 [27] discusses HTS compatibility scenarios with orbital separations of 2° and above. ITU-R S.524-9 [28] further specifies off-axis constraints for GEO earth stations at 30 GHz starting from an off-axis angle of 2°. In this context, the 3.5° adjacent-satellite angular separation considered in this paper falls within a realistic closely spaced GEO interference scenario (see Section VI-A).

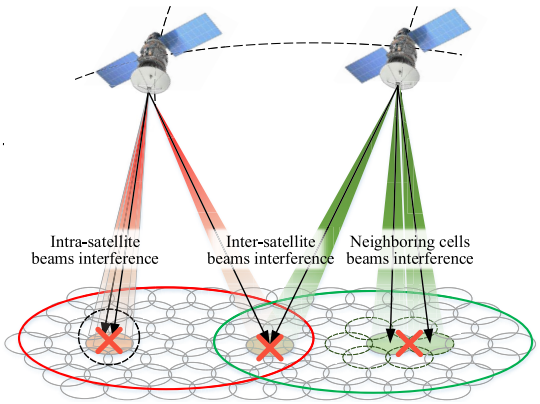


Fig. 3. The interference avoidance penalty mechanism in this research.

Meanwhile, modern GEO-HTS systems increasingly adopt multispot-beam architectures together with frequency reuse techniques to improve spectral efficiency and payload capacity [29]. Therefore, when adjacent GEO satellites employ co-frequency or highly reused spectrum over neighboring or overlapping service areas, CFI becomes a practical issue. This effect is particularly obvious for small-aperture user terminals with wide receive beams, which directly motivates the interference-avoidance mechanism introduced in this paper.

The penalty term is a mechanism used in RL to guide or restrict the agent's action. It is employed to penalize the agent when it exhibits unexpected actions, encouraging it to learn actions more aligned with expectations. To avoid potential issues such as beam interference and communication quality degradation, the penalty mechanism is introduced. Its specific usage will be presented in Section V, with this section focusing solely on its modelling. The interference avoidance scenarios considered in this research are illustrated in Fig. 3, where the penalty terms are defined as,

1) Intra-Satellite Beams Illuminate the Same Cell: If intra-satellite beams repeatedly illuminate the same cell, frequency overlap, signal superposition, etc., may cause beam interference. Therefore, each beam within the satellite should only be responsible for one specific cell within the visible area to avoid interference. Correspondingly, this penalty term is set to

$$\zeta_{s,t,1} = \sum_{v=1}^{N_s^{\mathsf{v}}} \underbrace{\left(\sum_{k=1}^{N_s^{\mathsf{k}}} \xi_{s,k,v,t} - 1 \right)}_{\text{Number of repeated illumination}} \times \kappa, \forall s \in \mathbb{S}, \tag{29}$$

where κ represents the penalty coefficient, multiplied by the number of repeat illuminate is the penalty amount.

2) Inter-Satellite Beams Illuminate the Same Cell: in the overlapping coverage of satellites, each cell should only be

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illuminated by one beam. Accordingly, this penalty term is

$$\zeta_{s,t,2} = \sum_{s'=1,s'\neq s}^{N^{\mathsf{s}}} \sum_{\forall v} \underbrace{\left(\sum_{k=1}^{N_s^{\mathsf{k}}} \xi_{s,k,v,t}\xi_{s',k',v,t} - 1 \right)}_{\text{Number of repeated illumination}} \times\kappa,$$
$$\forall s \in \mathbb{S}, \forall v \in \left\{\mathbb{V}_s \cap \mathbb{V}_{s'}\right\}, \forall k' \in \mathbb{K}_{s'}. \quad (30)$$

3) CFI From Neighboring Cells: As illustrated in Fig. 2, the various bandwidth reuse schemes chosen by beams from the same/different satellites may encompass the same sub-frequency bands, leading directly to the intra-/inter-satellite beam CFI. Therefore, it is imperative to avoid neighboring cells that contain the same frequency bands, thus reducing the possibility of interference. Likewise, we set the penalty term as

$$\zeta_{s,t,3} = \sum_{v=1}^{N_s^{\mathsf{v}}} \underbrace{\sum_{\forall v'} \left(\xi_{s,k,v,t}\xi_{s',k',v',t} \left(\sigma_{k',k,t} \neq 0 ? 1 : 0\right)\right)}_{\text{Number of CFIs in neighboring cells}} \times\kappa,$$
$$\forall v' \in \mathbb{N}_v, \forall s \text{ and } s' \in \mathbb{S}, \forall k \in \mathbb{K}_s, \forall k' \in \mathbb{K}_{s'}, \quad (31)$$

where $\mathbb{N}_v$ represents the neighborhood of v. It quantifies the degree of CFI of adjacent cells by monitoring the frequency band overlap status of the beams served by adjacent cells (triggered when $\sigma_{k',k,t} \neq 0$).

Therefore, combined with Eq. (29), Eq. (30), and Eq. (31), the total penalty term of satellite s at time slot t is as

$$\zeta_{s,t} = \zeta_{s,t,1} + \zeta_{s,t,2} + \zeta_{s,t,3}. \quad (32)$$

V. Multi-Agent Cooperative Decision-Making Based on QMIX

A. Motivation for the Adopted QMIX Model

According to discrete actions represented by Eq. (2) and Eq. (3), we employ the discrete action series RL models for optimization. Among them, the QMIX [30] model utilizes a centralized learning and distributed execution framework to address cooperative MADRL problems. Moreover, QMIX has been significantly upgraded based on deep Q-network (DQN) [31] and value-decomposition network (VDN) [32]. Specifically, QMIX not only inherits the advantages of DQN in approximating Q-value functions with deep neural networks (DNNs), enabling it to handle complex state and action spaces, but it also surpasses VDN by introducing a monotonicity constraint. This constraint ensures that QMIX can maintain the consistency between the individual agent Q-value function and the global Q-value function when decomposing and integrating the Q-value functions of multiple agents, thereby achieving the global optimal effect while ensuring the individual optimal strategy. Therefore, QMIX has become an ideal choice for dealing with this problem.

B. The Proposed Centralized Learning and Distributed Execution MADRL Framework

The proposed centralized learning and distributed execution framework based on QMIX for MSBH in the MBS constellation environment is shown in Fig. 4. For each beam of satellites, we deploy two agents, responsible for beam allocation decision-making and spectrum allocation decision-making, respectively. Different satellites share the same model, thus the total number of agents is $N^{\mathsf{k}} \times 2$. Correspondingly, the basic elements of each agent are configured as,

1) **State**: It is defined as the data traffic of all cells. Therefore, the state $\boldsymbol{s}_{s,k,t}$ for beam k of satellite s at time slot t is defined as

$$\boldsymbol{s}_{s,k,t} = \boldsymbol{D}_t. \quad (33)$$

Here, all agents share the same global state to facilitate collaborative optimization among multi-agents. It should be noted that the effective state of each satellite is $\boldsymbol{D}_t^s$.

2) **Action**: It is defined as the collaborative allocation of beams and spectrum. As mentioned before, there are a total of $N^{\mathsf{k}} \times 2$ agents in the system. The first N^{k} agents (defined as set $\mathbb{A}_1$) are responsible for beam allocation ($\boldsymbol{\Xi}_t$), and the final N^{k} agents (defined as set $\mathbb{A}_2$) are responsible for spectrum allocation ($\boldsymbol{\Phi}_t$). Therefore, assuming the i-th agent, the action $a_{s,k,t}$ for beam k of satellite s at time slot t is defined as

$$a_{s,k,t} = \begin{cases} \xi_{s,k,v,t}, & i \in \mathbb{A}_1, \\ \phi_{s,k,t}, & i \in \mathbb{A}_2. \end{cases} \quad (34)$$

3) **Reward**: We adopt the multi-objective coordinated and real-time optimization. Specifically, it is defined as the difference between the optimization goal and the penalty term, combining Eq. (28) and Eq. (32). Likewise, all agents share the same global reward to facilitate collaborative optimization among multi-agents. Therefore, the reward $r_{s,k,t}$ for beam k of satellite s at time slot t is

$$\begin{aligned} r_{s,k,t} &= \mathcal{F}_t - \sum_{s=1}^{N^{\mathsf{s}}} \zeta_{s,t}, \\ &= \alpha^{\mathcal{T}} \times \mathcal{T}^{\mathsf{lt}} - \beta^{\mathcal{D}} \times \mathcal{D}^{\mathsf{lt}} - \iota^{\mathcal{L}} \times \mathcal{L}^{\mathsf{lt}} - \sum_{s=1}^{N^{\mathsf{s}}} \zeta_{s,t}. \end{aligned} \quad (35)$$

This design ensures that agents are rewarded not only for improving throughput, delay fairness, and load balancing but also for avoiding inter-beam and co-frequency interference. Specifically, by maximizing $\mathcal{T}^{\mathsf{lt}}$ in $\mathcal{F}_t$, agents are incentivized to allocate beams and spectrum resources in ways that enhance overall system throughput; By minimizing $\mathcal{D}^{\mathsf{lt}}$ in $\mathcal{F}_t$ encourages agents to prioritize cells with higher queuing delays, improving delay fairness; By minimizing $\mathcal{L}^{\mathsf{lt}}$ in $\mathcal{F}_t$ pushes agents to distribute load evenly across satellites, avoiding overload on any single satellite; The penalty terms $\zeta_{s,t}$ directly discourage actions that lead to intra-satellite, inter-satellite, or co-frequency beam interference, improving system reliability and stability. Finally, the QMIX algorithm leverages its centralized learning and distributed execution framework, combined with the monotonicity constraint, to efficiently solve the joint optimization problem simultaneously represented by Eq. (18), Eq. (23), Eq. (27), and Eq. (32) within the reward function in Eq. (35), which will be described in detail below.

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TABLE II
THE GENERAL STRUCTURE OF THE AGENT NETWORK AND THE MIXING NETWORK

Network	Layer	Input×Output	Activation
Agent network	MLP	$(N_s^{\mathsf{v}} \times T + \lvert a_{s,k,t-1}\rvert) \times 256$	ReLU
	GRU	256×256	-
	MLP	$256 \times \lvert a_{s,k,t}\rvert$	-
Mixing network	Input layer	$N^{\mathsf{k}} \times 256$	ELU
	Output layer	256×1	-
Hyper-network	fc_1	$(N_s^{\mathsf{v}} \times T) \times (N^{\mathsf{k}} \times 256)$	Absolute
	fc_2	$(N_s^{\mathsf{v}} \times T) \times 256$	-
	fc_3	$(N_s^{\mathsf{v}} \times T) \times 256$	Absolute
	fc_4	$(N_s^{\mathsf{v}} \times T) \times 256$	ReLU
	fc_5	256×1	-

The network structure of the proposed framework is shown in Table II. 1) **The agent network**: Based on the QMIX paradigm, we deployed agent networks with shared identical structures. Furthermore, for each agent network the recurrent input consists of the current observed state $\boldsymbol{s}_{s,k,t}$ and the previous action $\xi_{s,k,t-1}$ (for beam allocation) or $\phi_{s,k,t-1}$ (for spectrum allocation), which are used to approximate the decomposed Q-value function for $Q_k\left(\boldsymbol{s}_{s,k,t}, \xi_{s,k,t}; \boldsymbol{\theta}_{s,k}^{\mathbf{B}}\right)$ or $Q_k\left(\boldsymbol{s}_{s,k,t}, \phi_{s,k,t}; \boldsymbol{\theta}_{s,k}^{\mathbf{S}}\right)$, where $\boldsymbol{\theta}_{s,k}^{\mathbf{B}}$ and $\boldsymbol{\theta}_{s,k}^{\mathbf{S}}$ represents the parameters of the agent network responsible for beam allocation and spectrum allocation, respectively. Moreover, it uses the ε-greedy method to select actions and empower the agent to balance between exploration and exploitation, as

$$a_{s,k,t} = \begin{cases} \arg\max\limits_{a_{s,k,t}} Q_k\left(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \boldsymbol{\theta}_{s,k}\right), & \text{with prob. } 1-\varepsilon, \\ \text{random action}, & \text{with prob. } \varepsilon, \end{cases} \tag{36}$$

where $\boldsymbol{\theta}_{s,k}$ represents the general term for $\boldsymbol{\theta}_{s,k}^{\mathbf{B}}$ and $\boldsymbol{\theta}_{s,k}^{\mathbf{S}}$. In addition, the exploration rate ε decreases linearly with training iterations, as

$$\varepsilon = \min\left\{0, \varepsilon - \frac{1}{N^{\mathsf{epo}}}\right\}, \tag{37}$$

where N^{epo} represents the number of training epochs. Moreover, each agent network comprises an Multi-Layer Perceptron (MLP) + Gated Recurrent Unit (GRU) + MLP architecture. Among them, the MLP consists of a single fully connected layer. In each agent network, the hidden layer dimension is set to 256 as well as the input dimension is the sum of the current observed state dimension and the action dimension, set to $(N_s^{\mathsf{v}} \times T + |a_{s,k,t-1}|) \times 256$. Here, $|a_{s,k,t-1}|$ represents the action dimension, where its value is defined as Eq. (42). Moreover,the GRU is specifically included to capture the temporal dependencies inherent in the traffic data. At each time slot t, the GRU unit takes the output $\boldsymbol{x}_t$ of the first MLP layer (which encodes the current observation $s_{s,k,t}$ and previous action $a_{s,k,t-1}$) as its input. Critically, it also incorporates the hidden state $\boldsymbol{h}_{t-1}$ from the previous time slot $t-1$, and uses its output $\boldsymbol{h}_t$ as an input for the next time slot. Among them, the GRU computes the new hidden state $\boldsymbol{h}_t$ using its internal gating mechanisms (reset and update gates), which can be formulated as

$$\boldsymbol{\upsilon}_t = \sigma(\boldsymbol{W}_\upsilon[\boldsymbol{h}_{t-1}, \boldsymbol{x}_t] + \boldsymbol{b}_\upsilon), //\texttt{Reset gate} \tag{38}$$

$$\boldsymbol{z}_t = \sigma(\boldsymbol{W}_z[\boldsymbol{h}_{t-1}, \boldsymbol{x}_t] + \boldsymbol{b}_z), //\texttt{Update gate} \tag{39}$$

$$\tilde{\boldsymbol{h}}_t = \tanh(\boldsymbol{W}_h[\boldsymbol{\upsilon}_t \odot \boldsymbol{h}_{t-1}, \boldsymbol{x}_t] + \boldsymbol{b}_h), \tag{40}$$

$$\boldsymbol{h}_t = (1 - \boldsymbol{z}_t) \odot \boldsymbol{h}_{t-1} + \boldsymbol{z}_t \odot \tilde{\boldsymbol{h}}_t, \tag{41}$$

where $\boldsymbol{\upsilon}_t$ and $\boldsymbol{z}_t$ represent the reset and update gates; $\tilde{\boldsymbol{h}}_t$ represents the candidate hidden state; σ represents the sigmoid function; $\odot$ represents the element-wise product; The terms $\boldsymbol{W}_\upsilon$, $\boldsymbol{W}_z$, $\boldsymbol{W}_h$ represent the learnable weight matrices, and $\boldsymbol{b}_\upsilon$, $\boldsymbol{b}_z$, $\boldsymbol{b}_h$ represent the corresponding learnable bias vectors for the respective gates and candidate state calculation. This updated hidden state $\boldsymbol{h}_t$ encapsulates a summary of the historical observation sequence and is passed to the final MLP layer to produce the Q-value. This mechanism allows each agent to make decisions not only based on the current state but also on the trajectory of past states and actions, which is crucial for learning effective long-term policies in a dynamic environment. Moreover, it should be noted that only the input MLP layer incorporates the ReLU activation function. Finally, the output dimension also corresponds to the action dimension $|a_{s,k,t}|$.

$$|a_{s,k,t}| = \begin{cases} N_s^{\mathsf{v}}, & a_{s,k,t} \in \mathbb{A}_1, \\ N_s^{\mathsf{reu}}, & a_{s,k,t} \in \mathbb{A}_2. \end{cases} \tag{42}$$

2) **The mixing network**: it is a feedforward neural network comprising two fully connected layers. It takes the outputs of each agent network as the input and combines them in a monotonic manner to produce the joint Q-value function, denoted as $\boldsymbol{Q}^{\mathsf{tot}}(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s)$. The monotonicity constraint of the mixing network ensures that the global Q value is a monotonic function of each local Q value, which means that an increase in the local Q value of a single agent will inevitably lead to an increase in the global Q value. This design ensures that the global strategy obtained during centralized training can be correctly decomposed into the optimal decision of each agent in decentralized execution, while reducing the complexity of multi-agent system optimization and improving training stability and synergy.

Lemma 1: The mixing network deployed in this work satisfies this monotonicity.

Proof: Please see Appendix. ■

3) **The hypernetwork**: It takes the state $\boldsymbol{s}_{s,t}$ as input and passes it through a series of fully-connected layers to produce a set of vectors. These output vectors are then reshaped to form the weights and biases parameters of the mixing network. The mapping relationship between the hypernetwork and the mixing network is as

- fc_1 layer mapping weight $\boldsymbol{W}_1$: the output of the first fully connected layer (fc_1) in the hypernetwork is reshaped into the weight matrix $\boldsymbol{W}_1$ of the input layer of the mixing network. The non-negativity constraint required by QMIX is enforced by applying an Absolute activation function to the layer generating this vector;

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- fc_2 layer mapping bias $\boldsymbol{b}_1$: the output of the second fully connected layer (fc_2) in the hypernetwork is reshaped into the bias $\boldsymbol{b}_1$ of the input layer of the mixing network;
- fc_3 layer mapping weight $\boldsymbol{W}_2$: the output of the third fully connected layer (fc_3) in the hypernetwork, also with an Absolute activation, is reshaped into the weight matrix $\boldsymbol{W}_2$ of the output layer of the mixing network;
- fc_4 and fc_5 layers mapping bias $\boldsymbol{b}_2$: the output of two-layer network (the fourth and fifth fully connected layer, fc_4 and fc_5) in the hypernetwork, with the first layer activated by ReLU, is reshaped into the bias $\boldsymbol{b}_2$ of the output layer of the mixing network.

Replay buffer: In each agent network, after action $a_{s,k,t}$ is taken, the reward $r_{s,k,t}$ and the next state $\boldsymbol{s}_{s,k,t+1}$ will be obtained. Then, the joint action $\boldsymbol{a}_{s,t} = \{a_{s,1,t}, \cdots, a_{s,N^{\mathsf{k}},t}\}$ and the next joint state $\boldsymbol{s}_{s,t+1} = \{\boldsymbol{s}_{s,1,t+1}, \cdots, \boldsymbol{s}_{s,N^{\mathsf{k}},t+1}\}$ can be obtained. These components are assembled into experience terms in the form of four-tuple $(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}, r_{s,t}, \boldsymbol{s}_{s,t+1})$. For simplicity, it is denoted as $(\boldsymbol{s}_t, \boldsymbol{a}_t, r_t, \boldsymbol{s}_{t+1})$, which is then stored in the replay buffer, denoted as

$$R = \{(\boldsymbol{s}_t, \boldsymbol{a}_t, r_t, \boldsymbol{s}_{t+1})_1, (\boldsymbol{s}_{t'}, \boldsymbol{a}_{t'}, r_{t'}, \boldsymbol{s}_{t'+1})_2, \cdots\}. \quad (43)$$

Target Q network: We employ the DQN [31] training approach to optimize the network parameters, wherein the target Q networks are established for each network to stabilize the learning process. It samples batches (m items) of experience items from the replay buffer through the experience replay mechanism, and estimates the target Q value of the next state, which will be used for training and updating of the Q network. For the experience $(\boldsymbol{s}_{s,k,t}, a_{s,k,t}, r_{s,k,t}, \boldsymbol{s}_{s,k,t+1})$, the target Q value is calculated as

$$\bar{Q} = \bar{Q}\left(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \bar{\boldsymbol{\theta}}\right) = r_{s,k,t} + \gamma \times \bar{Q}\left(\boldsymbol{s}_{s,k,t+1}, \arg\max_{a_{s,k,t}} Q\left(\boldsymbol{s}_{s,k,t+1}, a_{s,k,t}; \boldsymbol{\theta}\right); \bar{\boldsymbol{\theta}}\right), \quad (44)$$

where $\bar{\boldsymbol{\theta}}$ represents the parameter of the target Q network; γ represents the discount factor.

Optimization process: the following loss function is adopted, using the target Q network value as the estimated value and the Q value of the mixing network as the real value.

$$\mathcal{J}(\boldsymbol{\theta}) = \sum_{i=1}^{m} \left(\bar{\boldsymbol{Q}}^{\mathsf{tot}}\left(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \bar{\boldsymbol{\theta}}_s\right)_i - \boldsymbol{Q}^{\mathsf{tot}}\left(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s\right)_i\right)^2, \quad (45)$$

where $\boldsymbol{a}_{s,t} = \{a_{s,1,t}, \cdots, a_{s,k,t}, \cdots, a_{s,N^{\mathsf{k}},t}\}$. Moreover, $\bar{\boldsymbol{\theta}}$ is updated periodically (every C training steps) by copying parameters from the Q network $\left(\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}\right)$ to maintain synchronization.

Furthermore, the parameter update of the Q network is based on the gradient descent strategy, as follows

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \mu \times \nabla_{\boldsymbol{\theta}} \mathcal{J}(\boldsymbol{\theta}), \quad (46)$$

where μ represents the learning rate. Finally, the training algorithm flow utilized for optimization by the proposed framework is clearly shown in Algorithm 1.

Algorithm 1 The Training Algorithm Flow by the Proposed QMIX Framework.

Input: Training epochs N^{epo}; Current training iteration N^{iter} ; Batch size m; Target Q network update interval C; Current time slot t; More settings as Table III.
Data: Initialize parameters of all agent networks, mixing networks, hypernetworks and target Q networks; The environmental information as Section VI-A.
Output: Throughout $\mathcal{T}_t^{\mathsf{all}}$; Delay offset $\mathcal{D}_t^{\mathsf{all}}$; Load Offset $\mathcal{L}_t^{\mathsf{all}}$; Reward $r_{s,k,t}$; The network parameters.

▷ $N_{\mathsf{iter}} = 0$; $a_{s,k,0} = 0$; ▷ Random generate traffic arrival rate $\lambda_{s,k,t}$ of all cells; ▷ Initialize the state of ground cells;
while $N^{\mathsf{iter}} \leq N^{\mathsf{epo}}$ **do**
  ▷ Each agent obtains state $\boldsymbol{s}_{s,k,t}$ by Eq. (33) and previous action $a_{s,k,t-1}$;
  for *each agent in each satellite* **do**
    ▷ Agent network forward propagation obtains Q-value $Q\left(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \boldsymbol{\theta}_{s,k}\right)$ of action;
    ▷ Use ε-greedy method, select action $a_{s,k,t}$ by Eq. (36);
  end
  ▷ Mixing network collects Q-values $\{Q\left(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \boldsymbol{\theta}_{s,k}\right)\}$ of all agent networks, produce the joint Q-value $\boldsymbol{Q}^{\mathsf{tot}}(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s)$;
  ▷ Execute $\boldsymbol{a}_{s,t}$ by Eq. (34), transmit ground traffic;
  ▷ Calculate reward $r_{s,k,t}$ by Eq. (35);
  ▷ Generate new traffic according to $P(\lambda_{s,v,t+1})$ by Eq. (1), update new state $\boldsymbol{s}_{s,k,t+1}$ of cells;
  ▷ Obtain relevant indicator values $\boldsymbol{s}_{s,t}$, $\boldsymbol{s}_{s,t+1}$, $r_{s,t}$, $\mathcal{T}_t^{\mathsf{all}}$, $\mathcal{D}_t^{\mathsf{all}}$, $\mathcal{L}_t^{\mathsf{all}} = \mathsf{Step}(a_{s,k,t})$;
  ▷ Store experience $(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}, r_{s,t}, \boldsymbol{s}_{s,t+1})$ in relay buffer by Eq. (43);
  if $m \leq R$ **then**
    ▷ Sample batch m experience items;
    ▷ Calculated loss function value by Eq. (45);
    ▷ Update all network parameters $\boldsymbol{\theta}$ by Eq. (46);
    if $(N^{\mathsf{iter}} + 1)\%C == 0$ **then**
      ▷ Update all target networks parameters $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}$;
    end
  end
  ▷ Exploration rate ε decreases by Eq. (37);
  ▷ $\boldsymbol{s}_{s,k,t} = \boldsymbol{s}_{s,k,t+1}$; $a_{s,k,t-1} = a_{s,k,t}$; $N^{\mathsf{iter}} + +$;
  if $(N^{\mathsf{iter}} + 1)\%100 == 0$ **then**
    ▷ Change $\lambda_{s,v,t}$ of all cells;
  end
end

VI. Simulation Experiment and Result Analysis

A. Environment Simulation

The Viasat-1 satellite [6] is a high-throughput GEO satellite equipped with 72 spot beams (21 feed beams). The Viasat-3 constellation [33], representing Viasat's next generation of high-throughput Ka-band satellites, is anticipated to consist of 3 high-throughput GEO satellites. The Viasat communication system will also deploy hundreds of base stations equipped with small-diameter antennas on the ground. Based on these inspirations, we have simulated a MBS-GEO constellation-terrestrial integrated communication system operating in the Ka-band. A detailed configuration is presented in Table III.

Specifically, we deployed $N^{\mathsf{s}} = 3$ GEO satellites with an orbit altitude of $H = 36,000$ km [33]. Each satellite has $N^{\mathsf{k}} = 21$ beams, Ka-band downlink frequency $f^{\mathsf{ka}} = 20$ GHz [6], total power $P^{\mathsf{tot}} = 33$ dBW, maximum power

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TABLE III
SIMULATION ENVIRONMENT CONFIGURATION

Definition	Notation	Configuration
GEO satellites-related configuration		
Num of satellites	N^{s}	3
Number of beams for each satellite	N^{k}	21
Orbit altitude	H	36, 000 km
Ka-band downlink frequency	f^{ka}	20 GHz
Maximum transmit antenna gain	$G^{\mathsf{tran,max}}$	40.31 dBi
Total power of each satellite	P^{tot}	33 dBW
Maximum power limit of each beam	P^{max}	30 dBW
Transmit power of each beam	$\omega_{s,k}$	19.68 dBW
Number of bandwidth blocks	N^{sub}	5
Total bandwidth of each satellite	B^{tot}	1 GHz
Bandwidth of each block	B^{sub}	200 MHz
Ground-related configuration		
Number of cells	N^{c}	1, 071
Number of visible cells of each satellite	N^{v}	441
The packet's lifetime	T	30
Effective antenna aperture diameter	D	0.18 m
Receiving antenna gain of cell v	$G^{\mathsf{rec}}_{s,k,v}$	31.53 dBi
Traffic arrival rate	$\lambda_{s,v,t}$	$U[50, 500]$ Mbps
Wireless channel-related configuration		
3 dB beamwidth	Ψ^{bw}	1.5°
Off-axis angle when $G^{\mathsf{t}}_k(\Psi) = 0$ dBi	Ψ_1	12.23°
Near-in-side-lobe level	L_s	-30 dB
Signal propagation speed	c	3×10^8 m/s
Signal wavelength	λ	15 mm
Signal propagation path loss	P^{L}	209.59 dB
Noise power spectral density	N_0	-171.6 dBm/Hz
Model training-related configuration		
Number of training epoch	N^{epo}	2, 000
The exploration rate	ε	$0.9 \rightarrow 0$
The discount factor	γ	0.99
The learning rate	μ	0.001
Replay buffer size	R	20, 000
Batch size sampling from replay buffer	m	128
Penalty coefficient	κ	0.1
Target network update interval	C	200
Optimizer in Q network	-	Adam
Duration of each time slot	T^{slot}	2 ms

limit $P^{\mathsf{max}} = 30$ dBW [12], transmit power of each beam $\omega_{s,k} = P^{\mathsf{tot}} - 10\log_{10} N^{\mathsf{k}} = 19.78$ dBW, number of bandwidth blocks $N^{\mathsf{sub}} = 5$, total bandwidth $B^{\mathsf{tot}} = 1$ GHz [34], and bandwidth of each block $B^{\mathsf{sub}} = \frac{B^{\mathsf{tot}}}{N^{\mathsf{sub}}} = 200$ MHz.

Moreover, according to Eq. (9), the signal propagation path loss is calculated as $P^{\mathsf{L}} = 20 \times \log_{10}(H) + 20 \times \log_{10}(f^{\mathsf{ka}}) + 32.44 = 209.59$ dB. In addition, following the ITU-R S.672-4 [35], the transmit antenna gain for beam k of each satellite s at the off-axis angle Ψ is defined as

$$G^{\mathsf{tran}}_{s,k,v}(\Psi) = \begin{cases} G^{\mathsf{tran,max}}, & \Psi \le \Psi_0, \\ G^{\mathsf{tran,max}} - 3\left(\dfrac{\Psi}{\Psi_0}\right)^2, & \Psi_0 \le \Psi \le a\Psi_0, \\ G^{\mathsf{tran,max}} + L_s, & a\Psi_0 \le \Psi \le b\Psi_0, \\ G^{\mathsf{tran,max}} + L_s + 20 & \\ \qquad - 25\log\left(\dfrac{\Psi}{\Psi_0}\right), & b\Psi \le \Psi \le \Psi_1, \\ 0, & \Psi \ge \Psi_1, \end{cases} \tag{47}$$

where Ψ_0 denotes one-half the 3dB bandwidth ($\Psi^{bw} = 1.5^\circ$); Ψ_1 represents the off-axis angle Ψ when $G^{\mathsf{tran}}_{s,k,v}(\Psi) = 0$ dBi; $G^{\mathsf{tran,max}}$ represents the maximum transmit antenna gain on the main lobe, which is calculated as $G^{\mathsf{tran,max}} = 10 \times \log_{10}\left(4.93 \times \left(\frac{70}{\Psi^{bw}}\right)^2\right) = 40.31$ dBi; L_s represents the required near-in-side-lobe level for peak gain; a and b are constants. The utilization of antennas with smaller side lobes enables a more concentrated direction of signal energy toward the main beam, which is particularly advantageous in applications necessitating high precision pointing to specific regions and demanding interference avoidance. Therefore, the antenna radiation pattern with higher focusing ability (*i.e.*, $a = 3.16$, $b = 6.32$, and $L_s = -30$ dB) is chosen to ensure better transmission quality and reliability of satellite-terrestrial integrated communication, where $\Psi_1 = 12.23^\circ$.

TABLE IV
THE CORRESPONDING RELATIONSHIPS BETWEEN L_s, a, AND b, REFER TO [35]

a	b	L_s
2.58	6.32	-20 dB
2.88	6.32	-25 dB
3.16	6.32	-30 dB

The number of visible cells of each satellite is $N^{\mathsf{v}} = 441$. The constellation jointly serves $N^{\mathsf{c}} = 1071$ cells. All ground cell traffic follows the Poisson distribution with the traffic arrival rate $\lambda_{s,v,t}$. To simulate the non-uniform distribution of ground traffic, the values of $\lambda_{s,v,t}$ vary among different cells, which are randomly generated and belong to $50 - 500$ Mbps. Furthermore, the signal wavelength is calculated as $\lambda = \frac{c}{f^{\mathsf{ka}}} = \frac{3\times10^8 \text{ m/s}}{20 \text{ GHz}} = 15$ mm, where $c = 3 \times 10^8$ m/s represents the signal propagation speed. The ground receiving antenna is set as a Ka-band directional antenna, and its peak receiving gain is defined as $G^{\mathsf{rec}}_{s,k,v} = G^{\mathsf{rec,max}}_{s,k,v} = 10 \times \log_{10}\left(\frac{4\times\pi\times A}{\lambda^2}\right) = 31.53$ dBi, where $A = \frac{\pi \times D^2}{4}$ represents the effective aperture area of the antenna. Our analysis focuses on a critical and widespread terminal category, small-aperture user terminals, such as VSATs, where their typical characteristics are wider antenna beamwidth, making them particularly susceptible to interference from adjacent beams. Here, the effective aperture diameter D is set to $D = 0.18$ m. Following the ITU standard ITU-R S.465-6 [36], the receive antenna gain for beam k of each satellite s at the off-axis angle Ψ is defined as

$$G^{\mathsf{rec}}_{s,k,v}(\Psi) = \begin{cases} G^{\mathsf{rec,max}}_{s,k,v}, & \Psi \le \Psi^{\mathsf{min}}, \\ 32 - 25\log_{10}\Psi, & \Psi^{\mathsf{min}} < \Psi \le 48^\circ, \\ -10, & 48^\circ \le \Psi. \end{cases} \tag{48}$$

Furthermore, it is necessary to validate the practical relevance of this inter-satellite interference by analyzing the half-power beamwidth (HPBW). The HPBW, which defines the angular extent of the main-lobe region, is calculated as $70\frac{\lambda}{D} = 5.83^\circ$ for our terminal. In our simulation, the angular separation between adjacent satellites is between 3.49° and 3.56°. Therefore, this wide beamwidth (5.83°) is substantially larger than the angular separation in our simulated scenario.

To further highlight the practical relevance of the interference problem, we conducted a brief sensitivity analysis of the carrier-to-interference (C/I) ratio as a function of terminal aperture size and satellite angular separation. The C/I for a terminal pointed at the desired satellite and experiencing interference from an adjacent satellite (at $\Delta\Psi$ off-axis) can be calculated as $\text{C/I} = G^{\mathsf{rec,max}} - G^{\mathsf{rec}}(\Delta\Psi)$. We evaluated this for three representative terminal sizes, 0.18 m, 0.75 m, and 1.2 m. The resulting C/I values are shown in Table V. It can be observed that for the small-aperture terminal ($D = 0.18$ m), the C/I is 0 dB at small angular separations (3.5° and 5.0°).

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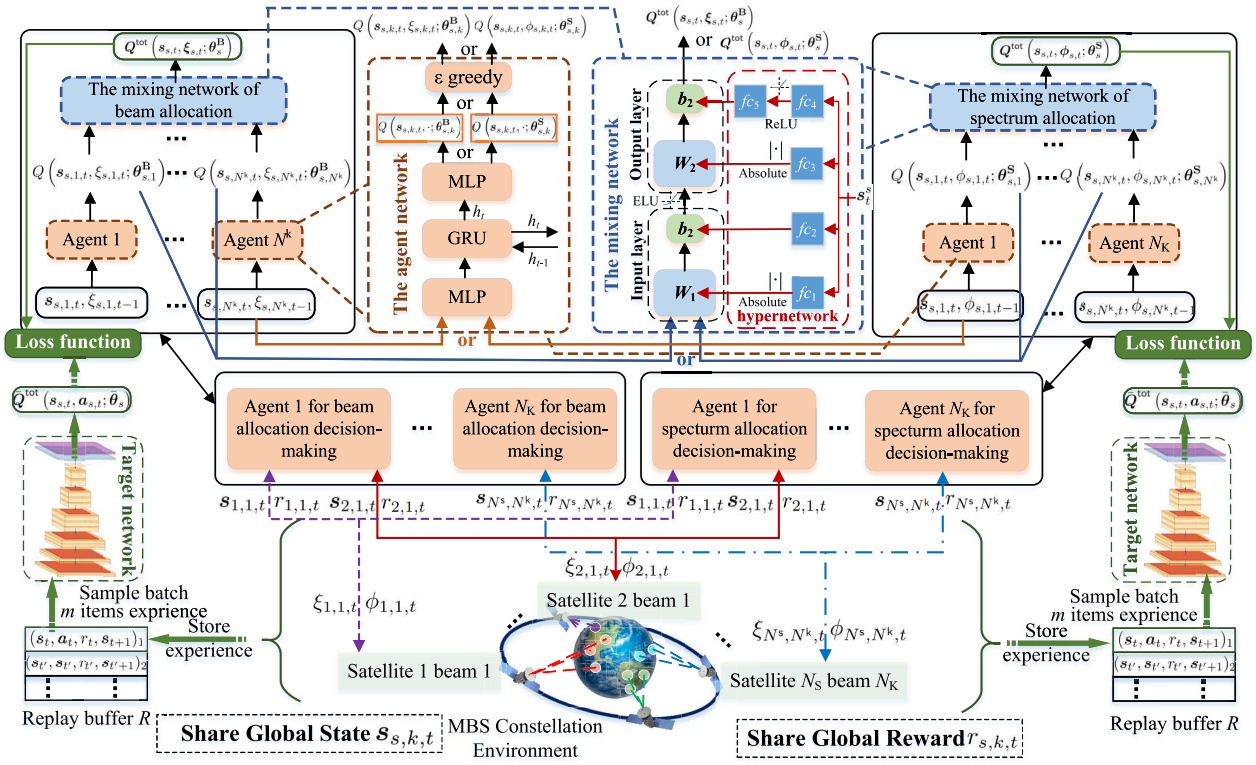


Fig. 4. The proposed multi-agent cooperative decision-making framework based on QMIX.

TABLE V
C/I (DB) FOR DIFFERENT ANGULAR SEPARATION AND TERMINAL SIZES

Angular Separation	3.5°	5.0°	10.0°
$D = 0.18$ m, HPBW $\approx 5.83°$ (our)	0	0	24.53
$D = 0.75$ m, HPBW $\approx 1.40°$	25.52	29.40	36.92
$D = 1.2$ m, HPBW $\approx 0.88°$	29.6	33.48	41.00

This confirms that the adjacent satellite's beam falls directly within the main lobe of the receiving antenna. An interference level of this magnitude would significantly degrade communication quality, thus necessitating avoidance measures. In sharp contrast, larger terminals with narrower beams provide sufficient attenuation to reduce the interference significantly. This analysis confirms that our work is of critical practical relevance and effectively addresses the challenging operational conditions faced by the growing market of small and mobile satellite terminals.

B. Training Performance and Ablation Comparison

During the training process, the relevant parameter settings are presented in Table III. The exploration rate varies from 0.9 to 0, encouraging the agent to take more random actions in the early iterations, exploring the environment to discover new possible strategies or knowledge. In the later stages of iterations, the agent is inclined to utilize known optimal strategies and knowledge. Fig. 5 illustrates the system reward value during the training epochs. At the beginning, due to the random initialization of the agent network, the mixing network, and the hypernetwork, and the lack of sufficient

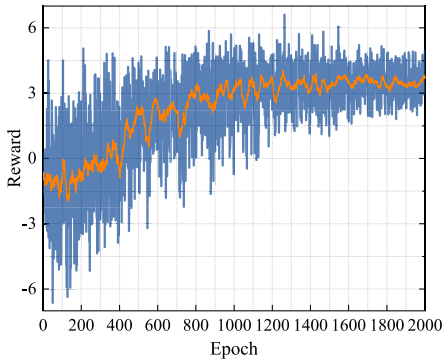
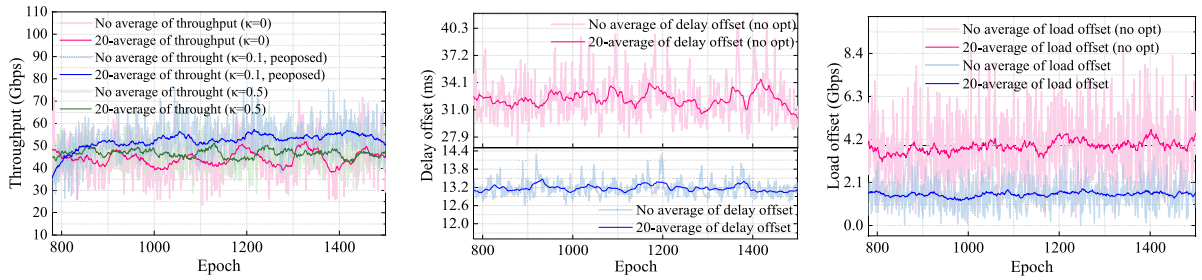


Fig. 5. Reward during the training epochs.

experience and knowledge, the reward values are relatively small. As exploration of the environment progresses, based on the experience replay mechanism, batches of data are sampled from the replay buffer to stabilize the learning process. Furthermore, the QMIX model learns the joint state-action Q-value function through a mixing network, optimizing overall system performance by leveraging the cooperative relationships among beam agents through centralized learning and distributed execution. It is evident that the proposed method converges rapidly during this process and tends to stabilize around 800 epochs. However, due to the greedy approach, the model still occasionally explores unknown actions with a small probability, resulting in small fluctuations in loss value in the subsequent phases.

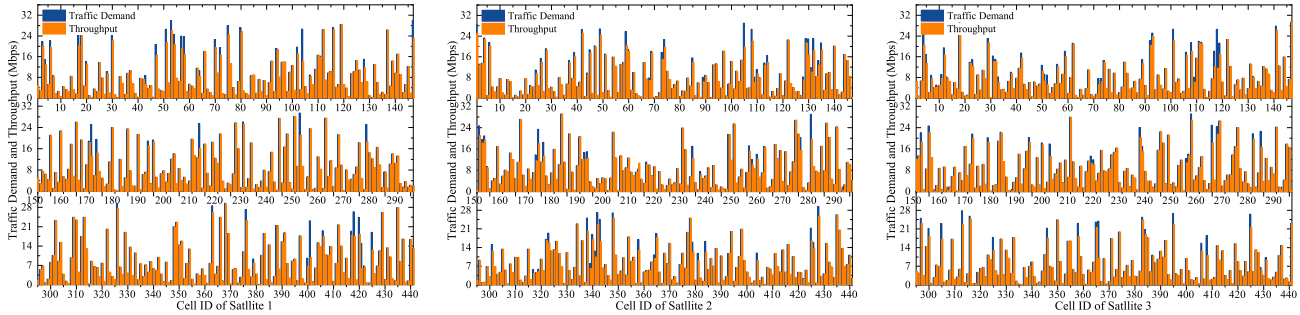
Moreover, the throughput, delay offset, and load offset during the stable training epochs are illustrated in Fig. 6. It can be

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(a) (b) (c)

Fig. 6. (a) Throughput, (b) Delay offset, (c) Load offset including the ablation comparison during the training stable epochs.



(a) (b) (c)

Fig. 7. (a) Traffic arrival rate $\lambda_t^{s,v}$ of all visible ground cells in one time slot of a satellite; Throughput and total traffic demands for part visible cells in (b) Satellite 1, (c) Satellite 2, (d) Satellite 3.

observed that the constellation throughput stabilizes at around 53 Gbps; the delay offset between cells stabilizes at around 13.2 ms; the load offset between satellites stabilizes at around 1.5 Gbps. In addition, in Fig. 6(a), Fig. 6(b), and Fig. 6(c), we set $\kappa = 0$, $\beta^{\mathcal{D}} = 0$, and $\iota^{\mathcal{L}} = 0$ respectively, to conduct an ablation comparison and verify the necessity and effectiveness of interference avoidance, delay fairness, and load balancing optimization. Specifically, according to Eq. (28) and Eq. (35), setting $\kappa = 0$ indicates that interference avoidance is not considered, setting $\beta^{\mathcal{D}} = 0$ indicates that delay fairness is not optimized, while $\iota^{\mathcal{L}} = 0$ indicates that load balancing is not optimized. In these scenarios, throughput, delay fairness, and load balancing perform unstable and poor. Without considering interference avoidance, the system capacity will decrease according to Eq. (4) and Eq. (6), which will in turn lead to a decrease in system throughput. It can be found that although without considering interference avoidance, its performance is obviously poor. Without optimizing delay fairness, the agent may overlook non-hotspot areas until their total traffic demands a value before they are illuminated, resulting in poor delay offset with an average of 32 ms. Similarly, if load balancing is not optimized, overlapping visible cells might be assigned to satellites with larger beam channel capacities to improve the system throughput, resulting in poor load offset with an average of 3.9 Gbps. In summary, the proposed method demonstrates good convergence and stability through the collaborative interaction among multi-agent.

Moreover, Fig. 6(a) also includes a sensitivity analysis of the penalty coefficient κ to validate its chosen value. Among them, the curve for $\kappa = 0$ represents the case without any interference penalty, where the agent does not learn to mitigate interference. As a result, its throughput converges to a lower and less stable state, demonstrating the necessity of the interference avoidance mechanism. On the other, we tested a high penalty value with $\kappa = 0.5$. In this scenario, the interference penalty term significantly dominates the overall reward signal. To maximize its reward, the agent is incentivized to adopt overly conservative strategies, strictly avoiding any action that might incur even a small penalty. This forces the agent to forgo many potentially high-throughput opportunities in favor of maintaining maximum spatial isolation between beams. As confirmed by the results, its throughput for $\kappa = 0.5$ converges to a level, which is significantly lower than that of our proposed setting ($\kappa = 0.1$). Therefore, the results validate that our selected value of $\kappa = 0.1$ strikes a robust and effective balance, enabling the agent to learn a policy that successfully avoids interference while still aggressively pursuing high-throughput performance.

C. Testing Performance

The corresponding relationship between the total traffic demand and throughput of the visible cells when illuminated by beams for each satellite are shown in Fig. 7(a), Fig. 7(b), and Fig. 7(c), respectively. It can be observed that the throughput of the majority of visible cells is very close to its total traffic demand. For each satellite s, define the average traffic satisfaction rate of all visible cells as

$$r_s = \frac{\sum_{v=1}^{N^{\mathrm{v}}} \frac{\boldsymbol{d}_{s,v,t}}{\mathcal{T}_{s,v,k,t}}}{N^{\mathrm{v}}}, \tag{49}$$

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TABLE VI

THE AVERAGE TRAFFIC SATISFACTION RATES OF ALL SATELLITES

r_s	Satellite 1	Satellite 2	Satellite 3
$r_s \geq 90\%$	92.5%	88.4%	89.1%
$r_s \geq 80\%$	95.4%	95.2%	95.4%
$r_s \geq 70\%$	98.1%	98.3%	98.6%

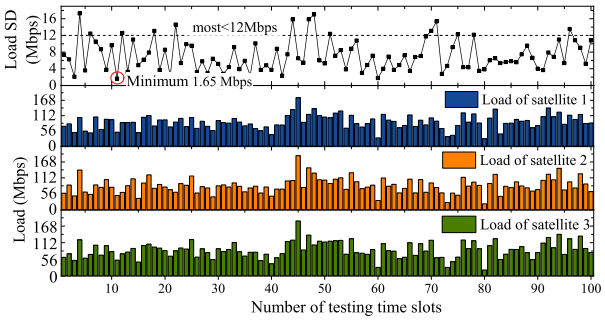


Fig. 8. Load and load SD of each satellite during testing time slots.

The average traffic satisfaction rates of all satellites are shown in Table VI. Therefore, it indicates that our proposed method enables the flexible allocation of constellation resources while ensuring the traffic demands of the majority of cells are met.

Moreover, Fig. 8 shows the load situation of each satellite during the 100-time slots in the testing phase. It illustrates the load balancing effect from two perspectives: spatial and temporal. Firstly, for spatial load balancing at each time slot, the top subplot presents the SD of the loads across the three satellites. It can be observed that the loads between different satellites are relatively balanced. For most cases, the load SDs are less than 12 Mbps. Secondly, for temporal load stability on each satellite, the bottom three subplots show the load trajectory of each individual satellite over time. While there are natural fluctuations due to the dynamic and non-uniform nature of ground traffic demands, the load on each satellite does not exhibit drastic, erratic swings. The loads remain within a stable operational range, demonstrating that our algorithm yields a consistent and predictable load distribution over time. The SD of each satellite's load over these 100 time slots is calculated to be 25.8 Mbps, 27.1 Mbps, and 27.2 Mbps respectively, which are all considered to be low, further confirming the long-term stability. This temporal stability, combined with the instantaneous spatial balance shown by the low SD, collectively demonstrates that our proposed method achieves effective and robust long-term load balancing for the multi-satellite system.

D. Comparative Performance With Other Works

Since the traffic arrival processes among different cells are mutually independent, and based on the independent additivity of the Poisson distribution, for a certain satellite s, the total traffic demand of its visible cells is $\sum_{v=1}^{N^{\mathrm{V}}} \lambda_{s,v,t}$. The baselines are shown in Table VII and the comparative throughput results are illustrated in Fig. 9, where the average throughput of multiple satellites is recorded for CMSBH. When the traffic demand is less than 2000 Mbps, the throughput differences among all

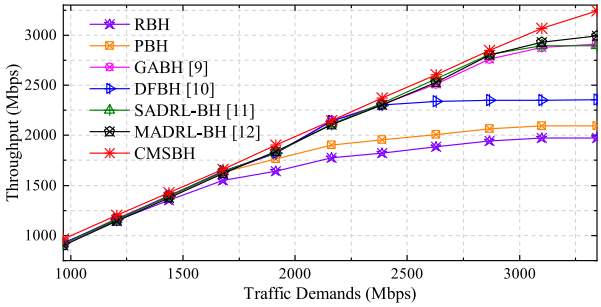


Fig. 9. Comparative performance in throughput.

TABLE VII

THE BASELINES USED IN THIS WORK

Baseline	Description
Random BH method (RBH)	The BHTP of each satellite is randomly generated, i.e., each satellite randomly illuminates N_{K} visible cells in each time slot. In addition, the spectrum resources of each beam for each satellite are also randomly selected from N^{s}_{reu} reuse schemes.
Periodic BH method (PBH)	For each satellite, all visible cells will be sequentially illuminated. For example, in the first time slot, illuminate cells No. 1-21(ID); in the second time slot, illuminate cells No. 22-42(ID); in the third time slot, illuminate cells No. 43-63(ID); execute in sequence. One cycle lasts $\frac{N_{\mathrm{C}}}{N_{\mathrm{K}}} = 21$ time slots.
BH method based on genetic algorithm (GABH) [9]	It is a classic BH method based on the meta-heuristic algorithm, which uses the GA algorithm to make decisions on BHTP and spectrum allocation schemes.
BH method based on delay fairness (DFBH) [10]	It is a classic BH method based on the heuristic algorithm, which uses the greedy algorithm to select the visible cell with the largest delay in each time slot to provide services.
BH method based on single-agent DRL (SADRL-BH) [11]	It treats a MBS as an agent with the optimization goals of minimizing ground traffic transmission delay and delay fairness as well as maximizing satellite throughput. Furthermore, it employed three separate DQN networks to independently optimize each objective.
BH method based on multi-agent DRL (MADRL-BH) [12]	It adopts the MARL algorithm and regards each beam of the satellite as an agent, which is responsible for the resource allocation and spatial pointing of the beam, with the optimization goals of minimizing delay fairness and maximizing satellite throughput.

algorithms are small, indicating that the satellite's capacity is sufficiently large to meet the ground traffic demand. However, as the traffic demand continues to increase, the throughput of the RBH and PBH algorithms declines significantly. This decrease is primarily due to RBH's random selection strategy and PBH's periodic sequential selection strategy, which fail to ensure that cells with higher traffic demands are adequately illuminated, leading to lower system throughput. For the DFBH algorithm, which prioritizes cells with the highest delay for illumination, system capacity increases to a certain extent, resulting in throughput performance significantly superior to RBH and PBH. Nevertheless, higher queuing delay does not necessarily correspond to greater stored traffic. Therefore, as the traffic demand continues to increase, its disadvantages become increasingly obvious, causing throughput to decline significantly. In contrast, the GABH, SADRL-BH, and MADRL-BH algorithms introduce optimization techniques that purposefully enhance system throughput by searching for global optima within the action space. This results in a notable increase in system throughput; however, these methods require extensive iterative searching, leading to high computational and time complexities, making them challenging to deploy on satellites with limited payload capacities.

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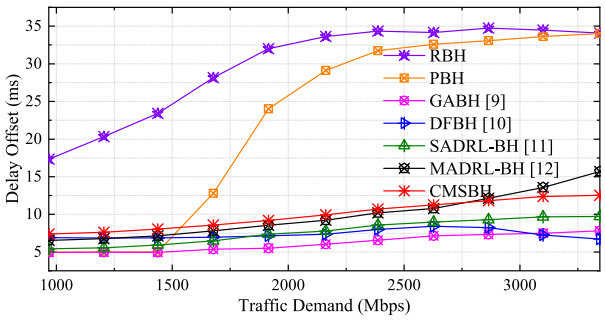


Fig. 10. Comparative performance in delay fairness.

Our proposed CMSBH algorithm demonstrates a clear advantage over the other methods. Notably, when addressing traffic demands of up to 3500 Mbps, throughput does not show a significant decline, indicating that system capacity can support even greater traffic demands. This is mainly attributable to the multi-agent system accumulating extensive experience in the replay buffer during training, allowing for the sharing of global experience. Moreover, the centralized learning and decentralized execution paradigm optimizes the utilization of collaboration and competition among agents. Furthermore, the proposed interference avoidance method significantly enhances system capacity by effectively avoiding inter-beam interference. In summary, CMSBH provides a more rational and efficient allocation of limited onboard resources.

The comparative delay fairness results are illustrated in Fig. 10. As traffic demand increases, the limited system capacity results in frequent cell traffic congestion, where illuminated beam fail to satisfy the stored cell traffic demand. Consequently, the delay offset among cells of all algorithms gradually increase. For the same reason, the RBH and PBH algorithms exhibit poor delay fairness, with PBH performing slightly better than RBH due to its periodic sequential selection strategy. Moreover, the DFBH algorithm, which prioritizes the illumination of cells with the maximum delay, achieves the best performance in the greater traffic demand. For the GABH, SADRL-BH, MADRL-BH, and our proposed CMSBH algorithms, the introduction of optimization techniques targeted at enhancing delay fairness leads to superior delay offset performance. Although our algorithm demonstrates slightly less favorable delay offset compared to GABH and SADRL-BH, its lower complexity makes it more suitable for deployment on satellites with constrained payload capacities. Furthermore, it is worth noting that CMSBH demonstrates stable performance even under varying traffic scenarios. This indicates that the model possesses strong scalability and generalizability for practical deployment.

E. Multi-Satellite BH Comparison

In addition, we also compared our proposed method with other MSBH methods, as outlined in Table VIII. Fig. 11(a) and Fig. 11(b) illustrate the average traffic satisfaction rate and normalized load offset for various MSBH strategies. Notably, the performance of RBH and PBH is suboptimal primarily because they do not account for traffic demands or satellite load conditions. In contrast, LBGT-MSBH and PH-MSBH first allocate cells based on satellite load, resulting in improved load offset. However, LBGT-MSBH employs a greedy algorithm that prompts each satellite to illuminate cells with the highest traffic demand, which does not necessarily ensure higher traffic satisfaction rates. Comparatively, PH-MSBH, which utilizes a mixed-integer linear programming approach for beam resource allocation, performs better than LBGT-MSBH. Nevertheless, the heuristic algorithm used in PH-MSBH leads to higher search complexity, and its staged optimization approach is less effective than the multi-objective collaborative optimization of CMSBH. Moreover, I-MSBH focuses solely on optimizing the BH of each satellite, achieving higher traffic satisfaction rates but neglecting inter-satellite load balancing and interference avoidance, resulting in less favorable load offset compared to CMSBH. Similarly, CMARL-MSBH treats each satellite as an

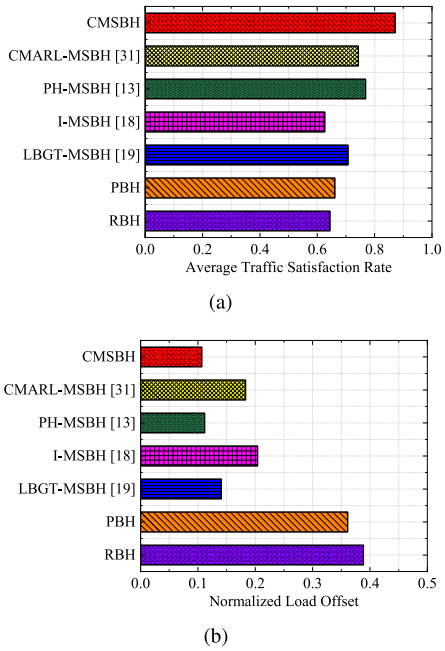


Fig. 11. Comparative performance in (a) average traffic satisfaction rate and (b) normalized load offset.

TABLE VIII
THE MSBH BASELINES USED IN THIS WORK

Baseline	Description
Load balancing and greedy-based throughput MSBH Method (LBGT-MSBH) [19]	This method allocates cells based on the satellite load, where each satellite employs a greedy algorithm to illuminate N_K visible cells with the highest traffic demand.
Independent MSBH method (I-MSBH) [18]	This method disregards load balancing and inter-beam interference avoidance, treating the multi-satellite BH process as multiple independent single-satellite BH operations, and determining the BH decisions for each satellite respectively.
Phased heuristic MSBH (PH-MSBH) [13]	This method sequentially decomposes the MSBH problem into three sub-problems: multi-satellite load balancing, single-satellite BH, and multi-satellite interference avoidance, and solves them respectively using heuristic solutions such as greedy algorithms.
Cooperative MARL MSBH (CMARL-MSBH) [37]	This method models each satellite as an agent and leverages a cooperative MARL algorithm and a CTDE framework to learn a cooperative beam allocation policy, thereby avoiding redundant coverage in overlapping areas and improving resource utilization.

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agent, avoiding overlapping beam coverage between different agents to improve system resource utilization, achieving a better traffic satisfaction rate than I-MSBH. However, due to the lack of explicit inter-satellite load balancing, its normalized load offset remains relatively poor. In contrast, the proposed CMSBH method consistently outperforms other algorithms, owing to its dynamic beam resource allocation based on traffic demands while simultaneously considering interference avoidance. This not only enhances system capacity but also improves throughput. Additionally, CMSBH employs a multi-objective collaborative optimization rather than the staged optimization strategy, which concurrently addresses load balancing during MSBH and dynamically allocates cells in real-time. This fundamental distinction from other algorithms contributes to its superior performance in the load offset indicator. Finally, it should be added that the proposed algorithm can be easily scaled to any number of satellites by adjusting the number of agents without redesigning the network structure.

VII. Conclusion

This paper presented a collaborative multi-satellite BH framework to address the dual challenges of potential satellite overload and beam interference. At the core of our approach, we formulated the dynamic resource allocation as a multi-objective optimization problem, designed to concurrently enhance system throughput, ensure delay fairness, and balance inter-satellite load. Moreover, we specifically developed a multi-level interference penalty mechanism that systematically accounts for intra-satellite, inter-satellite, and neighboring cell interference. To solve this complex optimization task, a centralized training and decentralized execution framework using the QMIX model was employed, yielding a globally coordinated policy. The effectiveness of the proposed method was validated through extensive simulations. Key findings indicate that our approach achieves a high traffic satisfaction rate, with the throughput of most cells closely matching their demand. Furthermore, the framework successfully maintains robust load balancing among satellites and ensures superior delay fairness compared to existing methods, demonstrating its practical value.

For future work, this framework could be extended to dynamic non-geostationary orbit (NGSO) constellations, presenting new challenges related to rapid topological changes and mobility management. Another valuable direction involves incorporating more sophisticated QoS metrics to support diverse communication services.

Appendix
Proof of Lemma 1

Monotonicity here refers to the consistency between the local Q-value function calculated by the distributed strategy and the joint Q-value function in "optimal performance", as follows

$$\arg\max_{\boldsymbol{a}_{s,t}} \boldsymbol{Q}^{\text{tot}}(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s)$$

$$= \begin{pmatrix} \arg\max_{a_{s,1,t}} Q_1(\boldsymbol{s}_{s,1,t}, a_{s,1,t}; \boldsymbol{\theta}_{s,1}) \\ \vdots \\ \arg\max_{a_{s,k,t}} Q_k(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \boldsymbol{\theta}_{s,k}) \\ \vdots \\ \arg\max_{a_{s,N^{\mathsf{k}},t}} Q_{N^{\mathsf{k}}}(\boldsymbol{s}_{s,N^{\mathsf{k}},t}, a_{s,N^{\mathsf{k}},t}; \boldsymbol{\theta}_{s,N^{\mathsf{k}}}) \end{pmatrix}, \tag{50a}$$

$$\text{s.t. } \frac{\partial \boldsymbol{Q}^{\text{tot}}(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s)}{\partial Q_k(\boldsymbol{s}_{s,k,t}, a_{s,k,t}; \boldsymbol{\theta}_{s,k})} \geq 0, \forall 1 \leq k \leq N^{\mathsf{k}}. \tag{50b}$$

The mixing network can be expressed as $\boldsymbol{Q}^{\text{tot}}(\boldsymbol{s}_{s,t}, \boldsymbol{a}_{s,t}; \boldsymbol{\theta}_s) = \boldsymbol{W_2}\text{ELU}(\boldsymbol{W_1 Q} + \boldsymbol{b_1}) + \boldsymbol{b_2}$, where $\boldsymbol{Q} = \left[Q_1(\boldsymbol{s}_{s,1,t}, a_{s,1,t}; \boldsymbol{\theta}_{s,1}), \cdots, Q_{N^{\mathsf{k}}}(\boldsymbol{s}_{s,N^{\mathsf{k}},t}, a_{s,N^{\mathsf{k}},t}; \boldsymbol{\theta}_{s,N^{\mathsf{k}}})\right]^{\mathrm{T}}$. We then have,

$$\begin{aligned} [fn1]\frac{\partial \boldsymbol{Q}^{\text{tot}}}{\partial \boldsymbol{Q}} &= \frac{\partial \text{ELU}(\boldsymbol{W_1 Q} + \boldsymbol{b_1})}{\partial \boldsymbol{Q}} \\ \boldsymbol{W}_2 &= \frac{\partial \text{ELU}(\boldsymbol{W_1 Q} + \boldsymbol{b_1})}{\partial (\boldsymbol{W_1 Q} + \boldsymbol{b_1})} \boldsymbol{W}_1 \boldsymbol{W}_2 \\ &\xlongequal{x = \boldsymbol{W_1 Q} + \boldsymbol{b_1}} \frac{\partial \text{ELU}(x)}{\partial x} \boldsymbol{W}_1 \boldsymbol{W}_2. \end{aligned} \tag{51}$$

As we know, $0 < \frac{\partial \text{ELU}(x)}{\partial x} \leq 1$. After passing through the absolute activation function of the hypernetwork, $\boldsymbol{W}_1 \geq 0$ and $\boldsymbol{W}_2 \geq 0$. Therefore, $\frac{\partial \boldsymbol{Q}^{\text{tot}}}{\partial \boldsymbol{Q}} \geq 0$, which satisfies Eq. (50b).

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